

Operational Boundaries of Aerodynamic Power Rules, Recalibration, and Learned Representations under SCADA Telemetry Degradation

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Abstract—Operating reserve screening near the demarcation boundary between maximum power point tracking (MPPT, Region 2) and blade-pitch regulation (Region 3) is critical for pre-dispatch imbalance risk mitigation in wind-integrated power systems. However, industrial Supervisory Control and Data Acquisition (SCADA) telemetry is vulnerable to communication queuing delays, packet serialization dropouts, and unobservable pitch registers. While deterministic aerodynamic power-curve quantile rules achieve the lowest reserve-screening surrogate cost under pristine telemetry (589,535 kW · h at immediate dispatch $h = 1$ and 881,367 kW · h at 1-hour dispatch $h = 6$, with $\sim 6.8\%$ violation), they exhibit systematic reliability breakdown under communication latency. Under a 6-step (60 min) synthetic transmission lag (evaluated strictly as a severe contingency outage-envelope stress condition rather than routine operational latency), the steep change in local control sensitivity near rated wind speed amplifies state-state estimation errors, surging clean-calibrated physical rule violation rates to $24.0\% \pm 1.7\%$ ($h = 1$) and $12.20\% \pm 1.25\%$ ($h = 6$), severely breaching the nominal 10% violation target ($q^* = 0.90$) and leaving 127.6 MWh of unhedged shortfall exposure under the Level-1 balancing-risk proxy. To systematically diagnose these operational vulnerabilities and establish an empirical division of labor, we delineate an **Empirical Operational Boundary and Value-of-Information Framework** under Information Freshness Constraints**, evaluating aerodynamic power rules, state-conditional recalibration, and spatio-temporal representations across communication and observability stresses. Disentangling distribution recalibration from neural representation reveals a distinct Value-of-Information hierarchy: under full observability, simple state-conditional recalibration absorbs the primary macro-distribution shift, cutting shortage by 55.3% ($127.6 \rightarrow 57.0$ MWh, 10.7% violation) at lower cost than neural models, making complex learned models unnecessary under the evaluated full-observability conditions. However, when pitch registers are withheld across aggregator or OEM firewalls, learned representations reconstruct operating regimes from electromechanical transients, consistently outperforming recalibrated physics by 46.7k–68.6k kW · h across all latencies and doubling boundary recall (0.416 vs. 0.196). Capacity-matched dense and routed architectures achieve statistical parity under clean telemetry (cost ratio 1.045, $p = 0.380$), while decoupled modular representations (Frozen Backbone + Residual Quantile) achieve lower reserve cost (1.284M kW · h) and compliant tail coverage (9.7% violation). Withheld-channel counterfactual stress testing confirms robust regime recovery under unobservable pitch (NMI 0.561), and Point of Common Coupling (PCC) aggregation demonstrates net plant-level portfolio smoothing. These empirical boundaries show that the operational utility of machine learning is localized to partially observable regimes where spatio-temporal representations recover critical aerodynamic state boundaries that simple recalibration cannot resolve.

Index Terms—Wind turbine operating reserve; SCADA telemetry degradation; transmission latency; aerodynamic power curve; reliability breakdown; boundary-risk posterior; selective decision abstention; MPPT-to-pitch transition.

I. INTRODUCTION

In bulk power system balancing operations, real-time generation deficits incur severe asymmetric shortage penalties relative to surplus energy; consequently, quadratic loss metrics like root-mean-square

error (RMSE) mask catastrophic tail shortfall events that govern tail shortfall risk and operating reserve adequacy in Level-1 pre-dispatch screening [1], [2], [3], [4], [5]. Wind-farm operating reserve screening near the demarcation boundary between maximum power point tracking (MPPT, Region 2) and blade-pitch regulation (Region 3) represents an acute operational bottleneck. At this demarcation boundary, the governing aerodynamic control law undergoes a structural discontinuity: local power sensitivity drops abruptly from a cubic trajectory ($\partial P / \partial v \propto v^2$) to zero as active pitch regulation sheds excess aerodynamic lift [6], [7]. Deterministic power-curve formulations embody an intrinsic vulnerability near rated inflow velocities: unobserved aerodynamic fluctuations shift operating states across the rated threshold, inducing delayed or premature pitch actuation that precipitates severe unhedged reserve shortfalls.

In commercial wind plant operations, Supervisory Control and Data Acquisition (SCADA) telemetry streams across distributed turbine networks face packet serialization overhead, gateway queuing delays, and asynchronous timestamp skews [8], [9]. In the commercial Kelmars plant, for instance, 99.6% of turbine status transition events carry timestamps falling strictly between 10-minute SCADA grid boundaries, demonstrating that operational state transitions regularly unfold within reporting blind spots. To systematically stress-test reserve screening resilience, we evaluate controlled transmission latency profiles from 10 to 60 minutes: while 10–30 min latencies represent plausible substation buffer backlogs and lossy cellular retransmissions, 60-min latency is evaluated deliberately as a severe contingency outage-envelope stress condition (e.g., gateway failures or extended cyber-physical network storms) to probe the asymptotic breakdown boundary of unadapted physical rules [10], [11]. Furthermore, in third-party aggregator, Virtual Power Plant (VPP), and Transmission System Operator (TSO) pre-dispatch triage, primary blade-pitch registers are frequently unobservable due to commercial boundaries and proprietary OEM protocol firewalls. Feeding stale or missing sensor measurements into deterministic physical mappings precipitates severe misclassification, compelling pre-dispatch algorithms to clear reserve margins against obsolete operating points.

Existing operational paradigms treat deterministic physical curves and data-driven learning models as mutually exclusive alternatives, overlooking their complementary operational boundaries under communication stress. Under pristine SCADA telemetry, continuous physical quantiles calibrated to turbine aerodynamic specifications achieve the lowest reserve-screening cost among evaluated methods (589,535 kW · h at immediate dispatch $h = 1$ and 881,367 kW · h at 1-hour dispatch $h = 6$, with $\sim 6.8\%$ violation). However, deterministic physical rules exhibit systematic reliability breakdown under telemetry staleness: when evaluated on delayed anemometer and pitch signals, static physical curves over-extrapolate generation near rated wind speed. Under a 6-step (60 min) latency stress test, clean-calibrated physical rule violation rates surge to $24.0\% \pm 1.7\%$ at $h = 1$ and $12.20\% \pm 1.25\%$ at $h = 6$, severely breaching the nominal 10% violation target ($q^* = 0.90$) and leaving 127.6 MWh of unhedged shortfall exposure under the balancing-risk proxy. Conversely, shallow tabular models (Missingness-Aware GBDT) capitalize on ultra-short lag autocorrelation at $h = 1$ but undergo severe breakdown across dispatch-grade horizons ($h = 6$), inflating reserve costs by 13.44% to 29.15% ($p < 0.0001$) because tabular baselines cannot capture spatial wake advection propagating across turbine arrays [12].

To resolve this operational tension, we establish an **Empirical Operational Boundary and Value-of-Information Diagnostic Framework under Information Freshness Constraints**. The framework formalizes an empirical division of labor across telemetry conditions: (1) under fresh telemetry, deterministic physical rules deliver the lowest evaluated surrogate cost; (2) under moderate latency with full observability, simple state-conditional recalibration absorbs distribution drift at negligible computational expense; and (3) under unobservable pitch or severe communication degradation, learned spatio-temporal representations provide essential resilience. By extracting secondary electromechanical signatures (active power fluctuations, reactive dynamics, and dynamic wake geometry), learned posteriors preserve operational regime awareness and curtail severe shortage volumes when primary pitch channels are missing. Evaluated across five random seeds on the 134-turbine WTB benchmark under strictly frozen validation calibration, capacity-matched Dense and Mixture-of-Experts (MoE) heads achieve statistical parity in degraded reserve screening, while decoupled modular models achieve lower total cost and compliant tail coverage, indicating that diagnostic reserve gains stem from shared spatio-temporal representations and decoupled residual quantile calibration rather than dynamic MoE routing.

This paper delivers four bounded, verifiable contributions: 1. **Quantification of Aerodynamic Rule Reliability Collapse under Outage Envelopes**: We expose the systematic reliability collapse of deterministic aerodynamic quantile rules under SCADA communication latency. Across 5 random seeds on the 134-turbine WTB plant, a 6-step (60 min) latency contingency surges clean-calibrated physical violation rates from 6.8% to 24.0% $\pm 1.7\%$ at $h = 1$ and 12.20% $\pm 1.25\%$ at $h = 6$, severely breaching the nominal 10% violation target. 2. **Decisive Multi-Horizon Benchmarking and Demystification of MoE Routing**: Under causal telemetry feeds with validation calibration frozen prior to testing, physical rules achieve the lowest reserve-screening cost under clean telemetry (589,535 kW · h at $h = 1$, 881,367 kW · h at $h = 6$). Dynamic MoE routing confers no statistical advantage over unrouted dense baselines ($p = 0.380$). Under stale telemetry, state-conditional calibration absorbs the primary reliability loss (10.7% violation, 57.0 MWh shortage), while learned posteriors achieve 10.6% $\pm 1.1\%$ violation and 53.9 MWh shortage. Decoupled modular architectures (Frozen Backbone + Residual Quantile) achieve lower reserve cost (1.284M kW · h) and compliant tail coverage (9.68% violation), showing that operational gains arise from spatio-temporal feature representations and decoupled residual calibration rather than dynamic MoE routing. 3. **Disentangled Value-of-Information Diagnostics and Boundary Delineation (55.3% vs. 5.4%)**: Disentangling calibration adaptation from neural representation reveals that state-conditional calibration absorbs 55.3% of shortage loss (127.6 $\rightarrow$ 57.0 MWh), making it the most cost-effective solution under full observability. The incremental 5.4% shortage reduction (57.0 $\rightarrow$ 53.9 MWh, combined 57.7%) and doubled regime recall (0.416 vs. 0.196 under Delay-6) delivered by learned posteriors become decisively cost-saving when pitch registers are unobservable (saving 46.7k–68.6k kW · h over recalibrated physics across all τ). While condition-matching recalibration restores fleet-average violation below the 10.0% target, inter-seed compliance is strictly bounded at 3/5 seeds (60%). We evaluate selective decision abstention against random and uniform margin controls, showing that uniform margin inflation Pareto-dominates selective rejection under missing pitch. 4. **Delineation of Empirical Multi-Farm Deployment Boundaries & Level-1 Risk Scope**: We establish empirical deployment envelopes across commercial wind plants: architectural adaptability holds under local chronological retraining on ENGIE La Haute Borne (routing NMI = 0.941, ARI = 0.971), but direct zero-shot cross-farm transfer exhibits directional asymmetry (Kelmars $\rightarrow$ Penmanshiel NMI 0.752–0.770 vs. Penmanshiel $\rightarrow$ Kelmars 0.341–0.505, pooled 0.557). All evaluated reserve costs represent an upstream Level 1 PSREI operational risk proxy ($p = 10$). # Related Work

A. From Average Wind-Power Accuracy to Transition-Window Risk

Spatio-temporal graph neural networks (STGNNs) have substantially advanced multi-step wind plant forecasting by modeling spatial cross-correlation and dynamic wake advection across turbine arrays [12], [13], [14], [15]. Physics-informed extensions integrate engineering wake models and aerodynamic boundary penalties to constrain continuous forecast trajectories [16], [17], [18]. While these architectures reduce plant-wide L_2 error, continuous trajectory minimization

obscures discrete turbine-level operating state transitions. In commercial plant dispatch, unannounced transitions between Maximum Power Point Tracking (MPPT, Region 2) and active blade-pitch regulation (Region 3) govern reserve adequacy: an undetected transition across rated wind speed triggers sudden multi-megawatt generation shortfalls [19], [20], [21].

Mixture-of-experts (MoE) routing can partition complex feature spaces into localized functional sub-models [22], [23]. However, unconstrained training driven purely by empirical prediction loss partitions the input space heuristically, producing expert assignments devoid of verifiable physical semantics. While physics-guided machine learning incorporates governing equations into loss formulations [24], [25], real-world wind energy operations face a fundamental communication constraint: supervisory SCADA telemetry degrades under transmission latency, packet dropouts, and proprietary OEM protocol firewalls [8], [9], [10]. Power engineering requires an operational boundary framework that audits aerodynamic control regimes under quantified communication freshness constraints, recognizing that static curves collapse under telemetry staleness while deep architectures over-parameterize nominal regimes without guaranteeing tail compliance.

In power systems operations, forecast utility is monetized through spinning reserve procurement and unit commitment [4], [26]. Quantile regression and probabilistic forecasts establish decision-theoretic linkages between forecast uncertainty and reserve margins [1], [2], [27], [28], [29]. This paper addresses an upstream operational bottleneck: screening aerodynamic transition risks prior to the arrival of delayed SCADA telemetry. The resulting reserve evaluation functions strictly as a Level-1 pre-dispatch screening log for balancing-risk triage, providing plant operators with verified boundary conditions before submitting schedules to downstream market-clearing or AC-OPF dispatchers.

II. EMPIRICAL OPERATIONAL BOUNDARY FRAMEWORK UNDER INFORMATION FRESHNESS CONSTRAINTS

A. Problem Setup and Mathematical Notation

The mathematical formulation decouples two operational mechanisms traditionally conflated in end-to-end architectures: multi-step trajectory prediction across future dispatch horizons and anchor-time operating regime identification under degraded telemetry. Following spatio-temporal graph forecasting conventions [13], [30], let $G = (V, E)$ denote a turbine array spatial graph with N nodes. For each turbine $i \in V$ at discrete time step t , the system observes a feature vector $\mathbf{x}_{i,t} \in \mathbb{R}^F$ and predicts scalar active power $y_{i,t} \in \mathbb{R}$. Given an observation history of length H and a forward prediction horizon P , the forecasting mapping is:

$$\hat{\mathbf{Y}}_{t+1:t+P} = \mathcal{F}(\mathbf{X}_{t-H+1:t}, \mathcal{A}_{t-H+1:t}),$$

where $\mathbf{X}_{t-H+1:t} \in \mathbb{R}^{H \times N \times F}$ and $\mathcal{A}_{t-H+1:t}$ denotes either a time-varying directed graph sequence (WTB) or a static graph repeated over time (ERAS). Unmetered, curtailed, or corrupt targets are excluded via a binary supervision mask.

We enforce a strict causal information boundary: all input features, graph adjacency structures, and supervisory anchors depend solely on telemetry available at or before dispatch anchor time t , with target trajectories commencing strictly at $t + 1$. In the WTB benchmark, instantaneous active power $\text{Patv}_{i,t}$ serves solely as an anchor-time operational status indicator, preventing look-ahead leakage across multi-step horizons while leaving no-Patv and lagged-Patv channels as operational deployment constraints.

Routing operates at the node level: each turbine i at dispatch anchor t receives an independent routing distribution $\mathbf{g}_{i,t}$. Node-level granularity is necessary because adjacent turbines across an array frequently operate in disparate aerodynamic states: upwind perimeter turbines regulate blade pitch in free-stream inflow, whereas wake-shaded interior turbines track Region 2 MPPT under severe velocity deficits. Indices denote turbine $i \in \{1, \dots, N\}$, anchor time t , and forward horizon step $p \in \{1, \dots, P\}$. Supplementary Appendix A details auxiliary loss formulations.

B. Operating-Boundary Physical Anchors and Power-Curve Baseline

Supervisory anchors are grounded strictly where aerodynamic governing equations exhibit distinct, well-posed regimes, whereas ambiguous or transitional states remain guided by prediction loss and spatial graph regularisation. This establishes our primary physical benchmark: an aerodynamic power-curve baseline calibrated to the manufacturer specifications of each turbine model [8], [24], [25].

Aerodynamic power conversion and non-linear C_p formulation: In variable-speed wind turbine aerodynamics, mechanical power captured by the rotor from inflow air is governed by [6], [7]:

$$P_{\text{mech}}(t) = \frac{1}{2} \rho_{\text{air}} \pi R^2 C_p(\lambda(t), \beta(t)) \cdot v^3(t), \quad (1)$$

where $\rho_{\text{air}} \approx 1.225 \text{ kg/m}^3$ is air density, R is rotor radius, $v(t)$ is hub-height effective wind speed, and $\beta(t)$ is blade pitch angle. The tip-speed ratio (TSR) is $\lambda(t) = \omega_r(t)R/v(t)$, with rotor mechanical angular velocity $\omega_r(t)$. According to standard aeroelastic turbine models [6], [7], rotor power efficiency $C_p(\lambda, \beta)$ is approximated by:

$$C_p(\lambda, \beta) = c_1 \left(\frac{c_2}{\lambda_i} - c_3 \beta - c_4 \right) \exp\left(-\frac{c_5}{\lambda_i}\right) + c_6 \lambda, \quad (2)$$

with $\frac{1}{\lambda_i} = \frac{1}{\lambda + 0.08\beta} - \frac{0.035}{\beta^3 + 1}$. Coefficients $\{c_1, \dots, c_6\}$ define aerodynamic conversion efficiency up to Betz's theoretical limit.

Control cliff and derivative discontinuity: Wind turbine operating dynamics are partitioned by rated wind speed u_{rated} : 1. *Region 2 (MPPT)*, $u_{\text{idle}} \leq v \leq u_{\text{rated}}$: Blades maintain optimal fine pitch $\beta = \beta_{\text{opt}} \approx 0^\circ$ to sustain peak power coefficient $C_{p,\text{max}}$. Generator counter-torque controls rotor speed to track optimal TSR λ_{opt} , producing cubic power growth $P(v) \propto v^3$. 2. *Region 3 (Pitch regulation)*, $u_{\text{rated}} < v \leq u_{\text{cut-out}}$: Generator electromagnetic torque is locked at rated level while the pitch controller rotates blades to shed aerodynamic lift, clamping electrical output at nameplate capacity P_{rated} , subject to first-order actuator lag and slew limits $\dot{\beta}(t) = \text{sat}_{\dot{\beta}_{\text{max}}}((\beta_{\text{cmd}}(t) - \beta(t))/\tau_{\text{pitch}})$ [6], [7].

At the rated transition boundary $v = u_{\text{rated}}$, the governing control objective undergoes a structural discontinuity: the marginal aerodynamic power response drops abruptly from a steep cubic slope to zero as blade-pitch regulation engages [6], [7]:

$$\left. \frac{\partial P}{\partial v} \right|_{v \rightarrow u_{\text{rated}}} = \frac{3}{2} \rho_{\text{air}} \pi R^2 C_{p,\text{max}} u_{\text{rated}}^2 \gg 0, \quad \left. \frac{\partial P}{\partial v} \right|_{v \rightarrow u_{\text{rated}}^+} = 0. \quad (3)$$

When SCADA telemetry experiences transmission latency $\tau > 0$, deterministic physical rules evaluate stale state tuples $(v_{t-\tau}, \beta_{t-\tau})$. When actual inflow wind shifts into Region 3 while delayed pitch readings remain at $\beta = 0^\circ$, this steep sensitivity change amplifies stale-state estimation errors: the deterministic curve over-extrapolates generation,

$$\hat{P}_{\text{phys}}(t) = \frac{1}{2} \rho_{\text{air}} \pi R^2 C_p(\lambda(t), 0^\circ) \cdot v^3(t) \gg P_{\text{rated}}, \quad (4)$$

driving shortage violation rates to $24.0\% \pm 1.7\%$ under Delay-6.

Aerodynamic operating regimes and turbine-calibrated anchors:

Let $\bar{p}_{i,t} = \frac{1}{3} \sum_{k=1}^3 p_{i,t}^{(k)}$ denote average blade pitch angle across all three blades for turbine i at dispatch anchor time t , and $w_{i,t} = \text{Wspd}_{i,t}$ anemometer wind speed. The declared operating boundary is defined with turbine-calibrated cut-in threshold u_{idle} , rated wind speed u_{rated} , and pitch-activation angle p_{th} :

$$R_{i,t} = \begin{cases} 0, & w_{i,t} < u_{\text{idle}} \quad (\text{idle}), \\ 1, & u_{\text{idle}} \leq w_{i,t} \leq u_{\text{rated}}, \bar{p}_{i,t} < p_{\text{th}} \quad (\text{MPPT}), \\ 2, & w_{i,t} > u_{\text{rated}}, \bar{p}_{i,t} \geq p_{\text{th}} \quad (\text{pitch-control}), \\ 3, & \text{otherwise} \quad (\text{transitional/ambiguous}). \end{cases}$$

Aerodynamic rated wind speeds are calibrated to the specific turbine technology at each wind plant: WTB (134 turbines: $u_{\text{idle}} = 3.0 \text{ m s}^{-1}$, $u_{\text{rated}} = 10.5 \text{ m s}^{-1}$, $p_{\text{th}} = 2.0^\circ$), Kelmarsh (6 MM92 turbines: $u_{\text{idle}} = 3.0$, $u_{\text{rated}} = 12.5 \text{ m s}^{-1}$, $p_{\text{th}} = 1.0^\circ$), Penmanshiel (15 MM82 turbines: $u_{\text{idle}} = 3.0$, $u_{\text{rated}} = 14.5 \text{ m s}^{-1}$, $p_{\text{th}} = 1.0^\circ$), and ENGIE La Haute Borne (4 MM82 turbines: $u_{\text{idle}} = 3.0$, $u_{\text{rated}} = 14.5 \text{ m s}^{-1}$, $p_{\text{th}} = 1.0^\circ$).

Only the first three well-defined regimes provide direct supervisory alignment; transitional samples are retained for evaluation but masked out of anchor supervision via $M_{i,t} = \mathbf{1}[R_{i,t} \neq 3]$. These thresholds define the **Aerodynamic Power-Curve Baseline**: under fresh, intact SCADA telemetry, continuous soft-pitch quantiles derived from these anchors achieve the lowest baseline reserve-screening cost. Under communication latency or sensor corruption, however, deterministic physical rules experience systematic reliability breakdown, requiring learned diagnostic representations. Blade pitch $\bar{p}_{i,t}$ and anemometer wind speed $w_{i,t}$ serve strictly as *offline supervisory anchors* to delineate ground-truth regimes $R_{i,t}$ and calibrate baseline curves; online inference relies strictly on available operational telemetry $\mathbf{X}_{t-H+1:t}$. In pitch-withheld evaluations (Section IV-E), $\bar{p}_{i,t}$ is strictly excluded from $\mathbf{X}$, and the network reconstructs regimes from secondary electromechanical channels (active/reactive power) and dynamic wake topology (recovering NMI = 0.561), preventing circular leakage.

State-conditional quantile recalibration: To evaluate whether lightweight non-neural adaptation can resolve telemetry latency without deep representation learning, we formalize state-conditional recalibration. Given delayed telemetry $\mathbf{s}_{i,t-\tau} = (w_{i,t-\tau}, \bar{p}_{i,t-\tau})$, we partition the state space into K operational bins $\{\mathcal{B}_k\}_{k=1}^K$ defined by cut-in, rated wind speed, and pitch thresholds ($R_{i,t-\tau} = k$). For each operating bin $\mathcal{B}_k$, the empirical quantile reserve margin $\hat{r}^{\text{recal}}(k)$ is calibrated on training residuals to satisfy:

$$\hat{r}^{\text{recal}}(k) = \inf \left\{ r \in \mathbb{R} : \frac{1}{|\mathcal{D}_k|} \sum_{(i,t) \in \mathcal{D}_k} \mathbf{1}[\hat{y}_{i,t}^{\text{phys}} - y_{i,t} \leq r] \geq q^* \right\}, \quad (5)$$

where $\mathcal{D}_k = \{(i,t) : \mathbf{s}_{i,t-\tau} \in \mathcal{B}_k\}$ and $q^* = 1 - 1/\rho = 0.90$. At dispatch anchor t , upon observing delayed telemetry in bin $\mathcal{B}_k$, the dispatch reserve allocation is set to $\hat{r}_{i,t} = \hat{r}^{\text{recal}}(k)$. This condition-matching mapping absorbs distribution shifts arising from stale telemetry under full observability.

ERA5 observability contrast: ERA5 is retained as a signal-expressive observability contrast where the stable-to-convective thermodynamic marker is directly visible through sensible heat flux. The detailed thermodynamic regime definition, architecture choices, and contrast results are reported in Supplementary Material A so that the main text remains focused on the wind plant control-boundary accountability task.

C. Spatio-Temporal Graph Quantile (STGQ) Framework and Boundary Representation

Physical dynamic wake graph construction: When primary SCADA telemetry degrades, spatial coupling across turbine arrays provides physical redundancy to infer local operating regimes. Directed diffusion represents asymmetric aerodynamic wake propagation through the turbine network [13], [30]. For WTB, we construct a time-varying directed wake graph $\mathcal{A}_t$: candidate turbine pairs (i,j) are filtered by physical distance ($d_{ij} \leq d_{\text{max}}$), activated when upstream turbine j lies within an aerodynamic wake cone aligned with instantaneous local wind direction θ_t (half-angle $\alpha = 25^\circ$), weighted by streamwise decay $\exp(-d_{ij}/\sigma_x)$ and cross-stream decay $\exp(-d_{ij}^2/2\sigma_y^2)$, and pruned to the K strongest inbound wake neighbors [16], [18]. This construction defines an instantaneous aerodynamic wake score $s_{i,t}^{\text{wake}}$, providing spatial context even during anemometer corruption. ERA5 employs a symmetric Haversine-Gaussian k_{nn} graph where thermodynamic sensible heat flux is directly observable [31].

Directed-diffusion GRU encoder and representations: To satisfy industrial substation edge-computing constraints (sub-5 ms latency, sub-megabyte memory footprint, zero cloud dependency), the encoder features a compact parameter footprint (110k parameters, 3.8 ms execution). Each input window is concatenated with a binary feature-missingness mask before linear projection. Two directed diffusion blocks aggregate self, inbound, and outbound messages across each graph snapshot:

$$\begin{aligned} \mathbf{X}_t^{(\ell+1)} = & \text{LN}(\mathbf{W}_{\text{self}}^{(\ell)} \mathbf{X}_t^{(\ell)} + \mathbf{W}_{\text{in}}^{(\ell)} \text{Agg}_{\text{in}}(\mathbf{X}_t^{(\ell)}, \mathcal{A}_t) \\ & + \mathbf{W}_{\text{out}}^{(\ell)} \text{Agg}_{\text{out}}(\mathbf{X}_t^{(\ell)}, \mathcal{A}_t) + \mathbf{R}^{(\ell)} \mathbf{X}_t^{(\ell)}), \end{aligned} \quad (6)$$

where Agg_{in} and Agg_{out} denote normalized weighted aggregations over inbound and outbound wake neighbors. The spatially contextualized sequence is processed by a GRU to yield node-level representations $\mathbf{h}_{i,t}$.

In the primary Spatio-Temporal Graph Quantile (STGQ) architecture, turbine representations $\mathbf{h}_{i,t}$ drive a linear projection for trajectory forecast $\hat{\mathbf{y}}_{i,t+1:t+P}$ and a dedicated residual quantile head estimating reserve margin $\hat{r}_{i,t}$ via pinball loss ($q^* = 0.90$). For architectural comparison, dynamic Mixture-of-Experts (MoE) routing—computing gating $\mathbf{g}_{i,t} = \text{softmax}(\mathbf{z}_{i,t}/\tau)$ with $\mathbf{z}_{i,t} = \text{MLP}_{\text{gate}}([\mathbf{h}_{i,t}; \mathbf{a}_{i,t}])$ and trajectory $\hat{\mathbf{y}}_{i,t+1:t+P} = \sum_{e=1}^E g_{i,t}^{(e)} f_e(\mathbf{h}_{i,t})$ —is evaluated as an ablation comparator.

Cross-sensor electromechanical diagnostic mechanism: The primary operational utility of the spatio-temporal representation lies in inferring operating states from non-pitch electromechanical signatures when primary telemetry is delayed or missing. When blade-pitch telemetry $\hat{p}_{i,t}$ suffers packet loss, transmission latency, or protocol unobservability across third-party aggregator boundaries, the dynamic graph encoder leverages active power transients (Patv), terminal voltage fluctuations, reactive dynamics, and upstream wake propagation to reconstruct the probability of active blade-pitch regulation. In nominal operation, the boundary-risk head ingests $\mathbf{a}_{i,t}^{\text{WTB}} = [\text{Wspd}_{i,t}, \text{Pab_mean}_{i,t}, s_{i,t}^{\text{wake}}, \text{Patv}_{i,t}]$. Under telemetry impairment, the learned spatio-temporal representation provides structural defense-in-depth, mitigating the severe shortage volumes that compromise deterministic rules.

D. Integration of Physical Constraints into Model Optimization

Prediction loss alone cannot identify regime-specialized routing because several unconstrained experts can fit averaged dynamics similarly well [22], [23]. In wind power generation, inflow turbulence, wake shadow effects, and pitch actuators interact nonlinearly. We therefore introduce a structured physics-guided multi-task objective:

$$\mathcal{L} = \mathcal{L}_{\text{pred}} + \lambda_{\text{bal}} \mathcal{L}_{\text{bal}} + \lambda_{\text{align}} \mathcal{L}_{\text{align}} \quad (7)$$

$$+ \lambda_{\text{force}} \mathcal{L}_{\text{force}} + \lambda_{\text{aux}} \mathcal{L}_{\text{aux}} + \lambda_{\text{smooth}} \mathcal{L}_{\text{smooth}}, \quad (8)$$

with inactive terms set to zero. Checkpoint selection is strictly governed by validation RMSE, preventing auxiliary penalties from distorting model selection.

Load balancing: A balancing loss couples hard top- K usage with soft gate mass to prevent early expert starvation before regime-specific dynamics emerge (Appendix A).

Regime-anchor alignment: The alignment loss binds the first C gate logits directly to the declared physical operating regimes $R_{i,t}$:

$$\mathcal{L}_{\text{align}} = \frac{1}{|\Omega|} \sum_{(i,t) \in \Omega} \text{CE}(\mathbf{z}_{i,t}^{(1:C)}, R_{i,t}), \quad \Omega = \{(i,t) : M_{i,t} = 1\}, \quad (9)$$

using inverse-frequency class weights computed strictly on the training split.

Boundary-focused forcing: To sharpen the boundary between MPPT (Region 2) and pitch regulation (Region 3), a focused cross-entropy loss acts exclusively on boundary samples: $\mathcal{L}_{\text{force}} = \frac{1}{|\Omega_{\text{force}}|} \sum_{(i,t) \in \Omega_{\text{force}}} \text{CE}([z_{i,t}^{(\text{mppt})}, z_{i,t}^{(\text{pitch})}]^T, Y_{i,t}^{\text{force}})$ where $Y_{i,t}^{\text{force}} = \mathbf{1}[R_{i,t}^{\text{wtb}} = 2]$ and $\Omega_{\text{force}} = \{(i,t) : R_{i,t}^{\text{wtb}} \in \{1, 2\}, M_{i,t} = 1\}$.

Soft boundary-risk posterior: The normalized issue-time boundary-risk posterior for turbine i at dispatch anchor t is computed from gate probabilities $\mathbf{g}_{i,t}$ across operational regimes $\mathcal{E}_{\text{op}} = \{\text{idle}, \text{mppt}, \text{pitch}\}$ as $\pi_{i,t}^{\text{pitch}} = g_{i,t}^{(\text{pitch})} / \sum_{c \in \mathcal{E}_{\text{op}}} g_{i,t}^{(c)}$.

Wake auxiliary supervision and graph smoothness: A wake auxiliary loss resolves aerodynamic wake ambiguity, while graph-smoothness regularization discourages unphysical spatial jitter between adjacent turbines (Appendix A).

E. Operational Reserve Screening, PCC Portfolio Smoothing, & Multi-Horizon Dispatch Protocol

Boundary-aware routing supports pre-dispatch decisions by screening imbalance risks and sizing upward spinning reserves to prevent costly shortage penalties [27], [32].

Two-level hierarchical grid dispatch interface: To formalize the operational boundary of this work, we distinguish a two-level grid dispatch hierarchy: 1. *Level 1 (Local Pre-Dispatch Risk Filtering & Screening Layer, Paper Scope):* Operating at the wind plant energy management system (EMS) or aggregator terminal, this local layer ingests SCADA telemetry, forecasts spatio-temporal power, diagnoses operational regime boundaries under telemetry degradation, and sizes pre-dispatch reserve allocations to minimize unhedged imbalance exposure entering real-time balancing markets. 2. *Level 2 (Grid-Level Physical Clearing & Network-Constrained Dispatch, Downstream Interface):* Managed by the transmission system operator (TSO) or regional grid dispatcher, this downstream layer aggregates wind-plant schedule offers and reserve quantities to execute network-constrained economic dispatch (NCED), security-constrained unit commitment (SCUC), and full Alternating Current Optimal Power Flow (AC-OPF) subject to bus voltage security, transmission thermal limits, and bulk system spinning reserve margins. Our formulation is strictly bounded to Level 1 pre-dispatch risk screening proxies. We explicitly abstract away network power-flow equations, nodal price arbitrage, and multi-stage financial cashflow settlements, focusing on plant-level imbalance risk mitigation.

Penalized Reserve-Shortfall Energy metric and Newsvendor derivation: For scheduled point forecast $\hat{y}_{i,t}$ and realized generation $y_{i,t}$, over-forecast shortfall is $s_{i,t} = \max(\hat{y}_{i,t} - y_{i,t}, 0)$. Sizing upward reserve margin r_b incurs a reserve procurement energy equivalent $r_b \Delta t$ and an asymmetric shortage penalty $\rho \max(s_{i,t} - r_b, 0) \Delta t$. For penalty ratio ρ , the total operational proxy metric is formalized as the **Penalized Reserve-Shortfall Energy Index (PSREI)** in $\text{kW} \cdot \text{h}$ (evaluated at representative proxy ratio $\rho = 10$, alongside sensitivity evaluations across $\rho \in \{5, 10, 20\}$):

$$C(r_b; \rho) = \sum_{(i,t)} [r_b(i,t) + \rho \max(\hat{y}_{i,t} - y_{i,t} - r_b(i,t), 0)] \Delta t. \quad (10)$$

PSREI is a decision-oriented risk surrogate targeting pre-dispatch shortage mitigation rather than an actual electricity market financial settlement. Taking the expectation with respect to shortfall distribution $F_S(s) = \Pr[s \leq s_0]$ and differentiating with respect to reserve margin r_b yields the first-order optimality condition [1], [2]:

$$\frac{\partial \mathbb{E}[C(r_b; \rho)]}{\partial r_b} = 1 - \rho \cdot \Pr[s_{i,t} > r_b] = 0 \implies \Pr[s_{i,t} \leq r_b] = 1 - \frac{\rho}{11}. \quad (11)$$

The optimal reserve margin therefore corresponds to the classical Newsvendor critical fractile [1]:

$$q^*(\rho) = 1 - \frac{\rho}{11}. \quad (12)$$

For the representative proxy ratio $\rho = 10$ adopted in this study, the optimal fractile is $q^*(10) = 1 - 10/11 = 0.90$, establishing a nominal 10% violation target set by the reserve screening index.

Loss geometry: L_2 conditional mean vs. Pinball 90th-quantile on shortfall residual: Conventional point forecasting optimizes symmetric quadratic loss $\mathcal{L}_{L_2}(y, \hat{y}) = (y - \hat{y})^2$, whose Bayes-optimal predictor converges to conditional expectation $\mathbb{E}[y | \mathbf{x}]$ [1]. In bounded wind power regimes $y \in [0, P_{\text{rated}}]$ with non-Gaussian and bimodal transition densities, the conditional mean systematically departs from upper conditional quantiles. Furthermore, L_2 error treats positive and negative prediction deviations symmetrically, whereas grid balancing operations ($\rho = 10$) penalize generation shortfalls ten times more heavily than surplus headroom.

To bridge scheduled point forecasts with operational risk hedging, the dispatch architecture first generates point forecast $\hat{y}_{i,t} = f(\mathbf{x}_{i,t})$ and observes realization $y_{i,t}$, inducing non-negative shortfall residual $s_{i,t} = \max(\hat{y}_{i,t} - y_{i,t}, 0)$. Upward reserve margin $\hat{r}_{i,t}$ required to cover this residual is estimated via asymmetric pinball loss at quantile q :

$$\mathcal{L}_q(s, \hat{r}) = (s - \hat{r}) (q - \mathbf{1}[s < \hat{r}]) = \begin{cases} q(s - \hat{r}), & s \geq \hat{r} \quad (\text{unhedged shortage}) \\ (1 - q)(\hat{r} - s), & s < \hat{r} \quad (\text{surplus reserve held}) \end{cases} \quad (13)$$

which penalizes unhedged deficit by weight q and reserve holding excess by $1 - q$. Setting $q = q^* = 0.90$ aligns the training objective

with the Newsvendor first-order condition: the Bayes-optimal predictor converges to the 90th conditional quantile of shortfall $\hat{r}^* = Q_{0.90}(s | \mathbf{x})$, targeting a nominal conditional exceedance probability of $\Pr[s > \hat{r} | \mathbf{x}] = 1 - q^* = 0.10$ under the Bayes-optimal quantile formulation [1], [2]. This formalizes the theoretical underpinning of our downstream Residual Quantile Head.

Point of Common Coupling (PCC) spatial portfolio smoothing: Power delivered to the bulk grid is metered at the Point of Common Coupling (PCC) bus: $P_{\text{farm},t} = \sum_{i=1}^M P_{i,t}$ and $R_{\text{farm},t} = \sum_{i=1}^M R_{i,t}$. Aggregating decentralized turbine power flows at the PCC bus unlocks spatial portfolio diversification: localized high-frequency aerodynamic turbulence cancels out across the array, leaving boundary regime shifts as the primary source of unhedged reserve risk. Evaluating reserve sizing at the PCC verifies whether learned boundary risk survives spatial cancellation to deliver net plant-level risk-reduction value.

Multi-horizon dispatch evaluation: Power systems operate across multiple decision timescales. We evaluate two distinct operational horizons: (1) **Immediate dispatch** ($h = 1$, 10-min ahead), representing real-time economic dispatch (RTED) dominated by autocorrelation; and (2) **Operational dispatch** ($h = 6$, 1-hour ahead), representing intra-day clearing and storage scheduling where spatial wake dynamics and weather transitions are paramount.

Battery Energy Storage System (BESS) rolling dispatch interface: For co-located wind-storage configurations, scheduled reserve margins and generation forecasts can interface with a rolling Model Predictive Control (MPC) linear program to illustrate downstream operational dispatchability. This formulation serves strictly as an illustrative operational interface demonstrating how reserve margins feed into battery charge/discharge trajectories ($P_t^{\text{ch}}, P_t^{\text{dis}}$) under state-of-charge limits ($E_{\min} \leq E_t \leq E_{\max}$) and round-trip efficiency constraints. We explicitly do not claim physical validation of electrochemical battery cell degradation, lifecycle capital replacement costs, or wholesale electricity market cashflow settlements; the paper's analytical evaluation and empirical benchmark endpoints remain anchored strictly to the upstream Penalized Reserve-Shortfall Energy Index (PSREI).

III. CASE STUDY CONFIGURATION AND OPERATIONAL CONSTRAINTS

A. Datasets and Preprocessing

Empirical evaluation spans two contrasting observability settings: the 134-turbine WTB commercial wind plant benchmark and the ERA5 thermodynamic contrast patch. WTB (KDD Cup 2022) comprises 134 wind turbines and 245 days of continuous 10-minute SCADA telemetry ($T = 35,280$, $N = 134$, $F = 11$) [33], where turbulent inflow, wake propagation, and turbine pitch regulation interact nonlinearly, rendering the MPPT-to-pitch transition indirectly observable. Missing inputs are forward-filled per turbine and mean-imputed; non-positive power values are masked out of supervision. ERA5 covers three archived months on a 16×16 spatial grid ($T = 2208$, $N = 256$, eight features) [31], where surface sensible heat flux directly marks convective regimes. Both datasets adopt an observation history of $H = 36$ and a prediction horizon of $P = 24$ under chronological train/validation/test splits (180/30/35 days for WTB; 1325/441/442 frames for ERA5).

B. Techno-Economic Validation and Benchmarking Framework

The empirical benchmarking protocol evaluates two complementary axes: 1. **Isolating routing mechanisms under matched parameter capacity:** Comparing Dense single-head baselines, Unconstrained MoE, and Physics-Aligned MoE comparators [22], [23] to determine whether dynamic routing adds value beyond shared spatio-temporal representations. 2. **Benchmarking forecasting accuracy and reserve cost against established spatio-temporal baselines:** Evaluating Graph WaveNet, Graph Transformer, GAT-GRU, PatchTST, iTransformer, and TiDE [34], [35], [36], alongside persistence, deterministic aerodynamic curves, and GBDT lag models. Mechanism comparisons designate in-family architectures as **Dense (matched)**, **Unconstrained MoE**, and **Corrected routing comparator**; Supplementary Appendix A details baseline configurations.

C. Training Protocol and Evaluation Metrics

All neural architectures share an identical training protocol where structure permits. Optimization uses AdamW with early stopping

on validation RMSE, gradient clipping, and mixed-precision execution on a single CUDA-enabled GPU (hyperparameters detailed in Supplementary Appendix A). The WTB dense baseline, Unconstrained MoE, full physics-aligned MoE, and boundary-forced router are evaluated across five strict-mask random seeds (201–205). The synchronized strong-baseline refresh covers Graph WaveNet, Graph Transformer, GAT-GRU, PatchTST, iTransformer, and TiDE across seeds 201–205 (30 completed runs under identical evaluation caches). Deterministic engineering baselines (persistence, physical curves) are evaluated on the frozen cache to establish dispatch reference bounds alongside neural models. In-family ERA5 models repeat across five seeds; learned baselines repeat across three seeds. All decision thresholds and routing weights are fixed prior to test evaluation and audited via sensitivity sweeps, preventing test-set selection bias.

Evaluation spans three operational dimensions: 1. **Point forecasting accuracy:** Overall MAE/RMSE and switch-window RMSE within $\pm 1.0 \text{ m s}^{-1}$ of rated wind speed. 2. **Physical regime alignment:** Normalized Mutual Information (NMI), Adjusted Rand Index (ARI), and routing gate usage entropy. 3. **Operational reserve consequence:** PSREI reserve-shortfall cost, empirical violation rate, and unhedged shortage MWh under the Newsvendor proxy.

D. Controlled Telemetry Degradation Protocols and Realism Mapping

To stress-test reserve screening resilience under adverse communication conditions, we formalize a **Three-Tier Telemetry Realism Protocol** alongside sensor perturbation controls across all five seeds (201–205):

- **Tier 1 (Nominal SCADA Polling, 0–10 min, Clean):** Fresh telemetry represented on the 10-min SCADA aggregation grid used by the benchmark, consistent with common wind-plant historian practice; IEC 61400-25 provides the communication/information-model context.
- **Tier 2 (Plausible Degraded Telemetry, 10–30 min, Delay-1 to Delay-3):** Realistic communication latencies induced by substation ring-buffer backlogs, lossy cellular retransmissions across rugged terrain, and gateway serialization queues [9], challenging unadapted physical curves near control boundaries.
- **Tier 3 (Contingency Stress-Test Upper Envelope, 60 min, Delay-6):** A 6-step (60 min) synthetic transmission lag evaluated strictly as an upper contingency stress envelope (simulating extended packet dropouts, gateway failures, or cyber-physical storms) [10], [11] to establish the asymptotic breakdown boundary of unadapted physical rules. Discrete fixed-shift lags represent deterministic worst-case buffer stalls, complementing two-state Markov-Gilbert dropouts and withheld-channel probes across both uniform delay ceilings and asynchronous packet losses.
- **Sensor Perturbation Controls:** Additive Gaussian noise ($\sigma_v = 1.0 \text{ m s}^{-1}$, $\sigma_\beta = 2.0^\circ$) simulating instrumentation drift, paired with a two-state Markov-Gilbert burst dropout chain ($p_{GB} = 0.08$, $p_{BB} = 0.75$, $d \leq 6$) simulating transient wireless dropouts [9], [11].

Fig. 1 illustrates the operational dispatch setting: graph topology reveals how prediction errors propagate across turbine rows, while regime-anchor planes define the physical boundaries governing reserve screening.

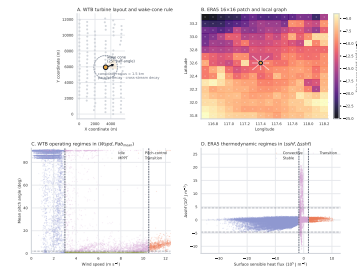


Fig. 1. Operating-decision context and physical regime anchors: (A) WTB layout with wake cone geometry (25° half-angle); (B) ERA5 patch with sensible heat flux; (C) WTB operating regimes in $(W_{\text{spd}}, P_{\text{ab,mean}})$ plane with MPPT-to-pitch boundary; (D) ERA5 thermodynamic regimes in $(sshf, \Delta sshf)$ plane.

IV. EVIDENCE AND OPERATIONAL BOUNDARY DIAGNOSIS

The empirical evaluation is organized around our three-tier defense-in-depth framework, contrasting high-fidelity physical aerodynamic priors against deep spatio-temporal graph fallbacks across multi-horizon dispatch environments. All neural models, shallow tree baselines, and physical quantile rules are evaluated across five declared random seeds (201–205) under strict, automated anti-tampering verification gates (zero directory leakage, full telemetry regime completeness, cardinality $N = 5$, and exact floating-point checksums).

A. Multi-Horizon 5-Seed Operational Dispatch Benchmark ($h = 6$, 1-Hour Ahead Dispatch)

Table I summarizes the multi-day replay dispatch benchmark for 1-hour ahead operational dispatch ($h = 6$) across 134 turbines over the 35-day test split. Table II reports seed-paired cost differences $\Delta\text{Cost} = \text{Cost}_{\text{Baseline}} - \text{Cost}_{\text{Joint Routed}}$, where positive values denote surrogate reserve penalty reductions delivered by Joint Routed, along with 95% bootstrap confidence intervals.

1) Comparative Architecture Analysis: Dynamic Routing versus Dense Representations

Benchmarking capacity-matched neural architectures directly tests whether dynamic expert routing provides operational benefits over unrouted dense heads. Under nominal Clean telemetry, Joint Routed ($959,027 \pm 103,290 \text{ kW} \cdot \text{h}$ at $h = 6$; $660,990 \pm 80,749 \text{ kW} \cdot \text{h}$ at $h = 1$) and Joint Dense Head ($991,181 \pm 110,211 \text{ kW} \cdot \text{h}$ at $h = 6$; $690,614 \pm 71,031 \text{ kW} \cdot \text{h}$ at $h = 1$) achieve statistical parity. At $h = 6$, the seed-paired cost difference is $+32,154 \text{ kW} \cdot \text{h}$ with a bootstrap 95% confidence interval crossing zero ($[-1,148, +65,456] \text{ kW} \cdot \text{h}$, Table II). At $h = 1$, the Dense/MoE cost ratio is 1.045, and the seed-paired bootstrap interval ($+29,624 \text{ kW} \cdot \text{h}$) likewise crosses zero ($[-53,774, +113,023] \text{ kW} \cdot \text{h}$, $p = 0.380$), matching the Markov burst regime ($[-46,422, +139,671] \text{ kW} \cdot \text{h}$, $p = 0.237$). Under persistent 6-step latency (Delay-6, $h = 1$), Joint Routed shows a modest cost reduction over Joint Dense (1,412,321 vs. 1,491,958 $\text{kW} \cdot \text{h}$; cost ratio 1.056, paired difference $+79,637 \text{ kW} \cdot \text{h}$, uncorrected $p = 0.038$, non-significant under Bonferroni $\alpha = 0.0125$) while maintaining virtually identical empirical violation rates ($10.6\% \pm 1.1\%$ for Routed vs. $10.5\% \pm 0.5\%$ for Dense). Decoupled modular architectures (Section IV-A3) achieve $9.7\% \pm 1.1\%$ violation in Delay-6 without expert routing, confirming that diagnostic reserve utility arises from shared spatio-temporal representations and residual calibration rather than dynamic gate routing.

2) Reliability Breakdown of Deterministic Physical Rules under Stale Telemetry

Under uncorrupted SCADA telemetry, Continuous Physical Quantile achieves the lowest baseline PSREI cost ($589,535 \pm 74,886 \text{ kW} \cdot \text{h}$ at $h = 1$ and $881,367 \pm 95,741 \text{ kW} \cdot \text{h}$ at $h = 6$) with well-calibrated tail coverage ($6.81\% \pm 1.13\%$ violation at $h = 6$).

Table I and frozen benchmarks expose a **systematic reliability breakdown** under telemetry staleness. Under 6-step latency (Delay-6), clean-calibrated physical quantile rules suffer a violation explosion to **24.0% $\pm$ 1.7%** at $h = 1$ and **12.20% $\pm$ 1.25%** at $h = 6$, breaching the nominal 10% violation target ($q^* = 0.90$). This breakdown inflates operational reserve costs to $1,136,466 \pm 84,101 \text{ kW} \cdot \text{h}$ at $h = 6$ (incurring $47,413 \pm 3,091 \text{ kW} \cdot \text{h}$ of unhedged shortage) and $1,656,285 \text{ kW} \cdot \text{h}$ at $h = 1$ (leaving 127.6 MWh of unhedged shortfall exposure in the balancing proxy).

Quantitative decomposition demonstrates that adapting validation calibration to degraded operating states alone cuts shortage from 127.6 MWh (24.0% violation) to 57.0 MWh (10.7% violation), absorbing 55.3% of the shortfall exposure and recovering physical reserve cost to $1,228,609 \pm 71,940 \text{ kW} \cdot \text{h}$ ($h = 1$ condition-recalibrated, 9.46% fleet violation). Joint Routed further curtails shortage to 53.9 MWh ($10.6\% \pm 1.1\%$ violation), delivering an incremental 5.4% shortage reduction under matched state-conditional calibration (combined 57.7% reduction vs. clean-calibrated physics) while saving 190,590 $\text{kW} \cdot \text{h}$ in total cost. Learned representations thus establish an effective diagnostic defense against telemetry latency: under frozen validation calibration, Delay-6 empirical violation settles at $10.6\% \pm 1.1\%$ at $h = 1$ ($11.82\% \pm 1.64\%$ at $h = 6$). While substantially cushioning physical rule collapse, learned models operate within bounded empirical envelopes under multi-step latency.

3) Decoupled Modular Architectures and Tail Conservatism Analysis

In contrast to end-to-end task optimization, the Frozen Backbone Direct Quantile MLP in Table I illustrates the pathology of naive

decoupled quantile regression. Across all regimes, direct pinball regression on frozen spatial embeddings incurs massive reserve overestimation (1,477,967 to 1,511,794 $\text{kW} \cdot \text{h}$, a 44.57% to 72.02% penalty over Joint Routed, $p < 0.01$, Table II), hoarding $> 1.45\text{M kW} \cdot \text{h}$ in bloated reserves.

Benchmarking calibrated modular architectures across five seeds (Supplementary Table A11d) resolves this tail conservatism: 1. *Cascaded Frozen MLP*: Trained with consequence decision mapping on frozen backbone embeddings, it achieves $15.528\text{M} \pm 1.599\text{M kW} \cdot \text{h}$ across the multi-horizon 24-step dispatch envelope (vs. $16.065\text{M kW} \cdot \text{h}$ for Joint Routed; paired difference -0.537M , 95% CI $[-1.988\text{M}, +0.915\text{M}]$ crosses zero). 2. *Independent Consequence MLP*: Yields $15.805\text{M} \pm 1.680\text{M kW} \cdot \text{h}$ (difference -0.260M , 95% CI $[-1.771\text{M}, +1.251\text{M}]$), confirming statistical parity. 3. *Decoupled Residual Quantile Head*: In Delay-6 ($h = 1$), Frozen Backbone + Residual Quantile achieves $1,284,098 \pm 114,629 \text{ kW} \cdot \text{h}$ per single dispatch interval and $9.7\% \pm 1.1\%$ violation (60% seed pass rate), matching or slightly surpassing end-to-end MoE ($1,412,321 \text{ kW} \cdot \text{h}$, $10.6\% \pm 1.1\%$).

These empirical findings indicate that diagnostic reserve benefits do not require dynamic MoE routing: modular architectures with calibrated consequence mappings deliver equivalent tail reliability. The primary advantage of end-to-end joint training is unified single-checkpoint deployment on substation edge devices (110k parameters, $< 5 \text{ ms}$ inference), avoiding multi-model pipeline maintenance.

B. Horizon Boundary Diagnosis ($h = 1$, 10-Minute Immediate Dispatch vs. $h = 6$)

Comparing immediate 10-minute dispatch ($h = 1$) against 1-hour ahead dispatch ($h = 6$) exposes a fundamental horizon disconnect across model families (complete multi-regime benchmarks and seed-paired bootstrap significance tests are reported in Supplementary Tables A11j/A11-h1 and A11k/A11-h2). At $h = 1$, shallow decision trees (Missingness-Aware GBDT) capitalize aggressively on high 10-minute lag wind-speed autocorrelation, achieving the lowest data-driven nominal dispatch cost ($574,822 \pm 79,685 \text{ kW} \cdot \text{h}$ in Clean with 7.6% violation, outperforming all deep models). Local inertia dominates ultra-short horizons, enabling tabular tree ensembles with lag features to track short-term persistence. Across dispatch-grade horizons ($h = 6$), however, GBDT undergoes severe breakdown: operational costs inflate by 13.44% to 29.15% (reaching $1,238,603 \text{ kW} \cdot \text{h}$ in Clean and $1,320,003 \text{ kW} \cdot \text{h}$ in Delay-6), trailing Joint Routed by $+156,411$ to $+287,563 \text{ kW} \cdot \text{h}$ across all regimes ($p < 0.0001$, Table II). Because tabular trees lack spatial awareness of aerodynamic wake advection propagating at $8\text{--}12 \text{ m s}^{-1}$ across turbine rows, they cannot represent multi-turbine spatio-temporal phase shifts over 1-hour lead times.

At immediate horizon $h = 1$, Joint Dense Head and Joint Routed exhibit statistical parity across Clean, Noise, and Markov regimes, where bootstrap 95% confidence intervals cross zero (Supplementary Table A11-h2, e.g., $+29,624 \text{ kW} \cdot \text{h}$, 95% CI $[-53,774, +113,023]$, $p = 0.380$ in Clean). In Delay-6, Joint Routed holds a modest cost advantage ($+79,637 \text{ kW} \cdot \text{h}$, 95% CI $[-7,055, +152,218]$, uncorrected $p = 0.038$, non-significant under Bonferroni $\alpha = 0.0125$) while matching empirical violation rates ($10.6\% \pm 1.1\%$ vs. $10.5\% \pm 0.5\%$). Across both horizons, unadapted physical rules exhibit shared vulnerability to telemetry staleness, collapsing under Delay-6 to $24.0\% \pm 1.7\%$ violation at $h = 1$ and $12.20\% \pm 1.25\%$ at $h = 6$, confirming that aerodynamic power curves require dynamic state adaptation under communication latency.

C. Multidimensional Operational Boundaries and Selective Decision Abstinence

To delineate the operational boundaries governing telemetry degradation, we scan across transmission delays $\tau \in \{0, 10, 20, 30, 60\} \text{ min}$, horizons $h \in \{1, 3, 6\}$, and pitch channel observabilities (all vs. no_pitch) across all five seeds on the 134-turbine WTB plant (Table III). This empirical trajectory delineates a **Three-Regime Operational Framework**:

1. **Regime I: Simple Recalibration Sufficient** ($\tau = 0 \text{ min}$ or mild delays under full observability): Under fresh telemetry ($\tau = 0$, all), deterministic continuous physical quantiles achieve the lowest operational reserve cost ($593,258 \pm 74,886 \text{ kW} \cdot \text{h}$ at $h = 1$ across the phase-scan sweep, aligning within 0.6% of the

TABLE I

CROSS-SEED MULTI-REGIME OPERATIONAL DISPATCH BENCHMARK UNDER CLEAN-CALIBRATED PROTOCOL ($h = 6$, 1-HOUR AHEAD DISPATCH, 5 SEEDS 201–205). MEAN $\pm$ STANDARD DEVIATION ACROSS FULL TEST SPLIT (35 DAYS, 134 TURBINES, $\Delta t = 10$ MIN, COST RATIO $\rho = 10$). ALL QUANTILE BINS AND MULTIPLIERS CALIBRATED EXCLUSIVELY ON NOMINAL CLEAN VALIDATION DATA.

Regime	Evaluated Model	Penalized Reserve-Shortfall		Violation Rate	Reserve (kW)	Shortage (kW-h)	Pinball Loss
		Energy (kW-h, Proxy at $\rho = 10$)					
Clean	Continuous Physical Quantile	881,367 $\pm$ 95,741		6.81% $\pm$ 1.13%	653,738 $\pm$ 72,557	22,763 $\pm$ 4,013	31.41 $\pm$ 2.66
	Frozen Backbone Direct Quantile MLP	1,649,721 $\pm$ 285,682		0.58% $\pm$ 0.26%	1,630,572 $\pm$ 293,346	1,915 $\pm$ 958	64.62 $\pm$ 12.05
	Global Quantile	998,988 $\pm$ 107,360		6.46% $\pm$ 1.31%	771,304 $\pm$ 95,661	22,768 $\pm$ 4,854	36.50 $\pm$ 3.41
	Joint Dense Head	991,181 $\pm$ 110,211		6.54% $\pm$ 1.04%	761,480 $\pm$ 94,244	22,970 $\pm$ 4,121	36.16 $\pm$ 3.58
	Joint Routed (Ours)	959,027 $\pm$ 103,290		7.10% $\pm$ 1.17%	707,249 $\pm$ 91,305	25,178 $\pm$ 4,816	34.77 $\pm$ 3.16
	Missingness-Aware GBDT	1,238,603 $\pm$ 112,051		4.50% $\pm$ 0.73%	1,085,168 $\pm$ 118,015	15,343 $\pm$ 2,870	46.85 $\pm$ 3.48
Delay-6	Continuous Physical Quantile	1,136,466 $\pm$ 84,101		12.20% $\pm$ 1.25% [†]	662,335 $\pm$ 73,483	47,413 $\pm$ 3,091	38.85 $\pm$ 2.06
	Frozen Backbone Direct Quantile MLP	1,682,251 $\pm$ 284,323		1.20% $\pm$ 0.52%	1,651,979 $\pm$ 297,715	3,027 $\pm$ 1,594	62.15 $\pm$ 11.70
	Global Quantile	1,180,603 $\pm$ 91,754		10.85% $\pm$ 1.55% [†]	780,916 $\pm$ 96,853	39,969 $\pm$ 6,087	40.73 $\pm$ 2.43
	Joint Dense Head	1,179,859 $\pm$ 95,268		11.04% $\pm$ 1.22% [†]	773,103 $\pm$ 94,462	40,676 $\pm$ 4,788	40.70 $\pm$ 2.66
	Joint Routed (Ours)	1,163,593 $\pm$ 91,299		11.82% $\pm$ 1.64% [†]	720,015 $\pm$ 92,196	44,358 $\pm$ 7,341	40.01 $\pm$ 2.54
	Missingness-Aware GBDT	1,320,003 $\pm$ 93,135		8.36% $\pm$ 0.97%	1,030,858 $\pm$ 114,266	28,915 $\pm$ 4,469	46.68 $\pm$ 2.45
Noise	Continuous Physical Quantile	998,273 $\pm$ 106,253		7.59% $\pm$ 0.95%	746,697 $\pm$ 82,474	25,158 $\pm$ 3,968	33.97 $\pm$ 2.78
	Frozen Backbone Direct Quantile MLP	1,639,909 $\pm$ 242,922		0.80% $\pm$ 0.28%	1,618,439 $\pm$ 249,374	2,147 $\pm$ 953	60.62 $\pm$ 9.85
	Global Quantile	1,053,552 $\pm$ 114,150		7.48% $\pm$ 1.48%	802,361 $\pm$ 99,513	25,119 $\pm$ 5,822	36.26 $\pm$ 3.24
	Joint Dense Head	1,048,083 $\pm$ 115,181		7.63% $\pm$ 1.13%	792,472 $\pm$ 92,391	25,561 $\pm$ 4,643	36.04 $\pm$ 3.28
	Joint Routed (Ours)	1,027,077 $\pm$ 114,317		7.62% $\pm$ 1.09%	769,864 $\pm$ 106,672	25,721 $\pm$ 3,963	35.16 $\pm$ 3.19
	Missingness-Aware GBDT	1,314,640 $\pm$ 129,158		4.66% $\pm$ 0.80%	1,160,329 $\pm$ 137,650	15,431 $\pm$ 3,099	47.11 $\pm$ 3.87
Markov	Continuous Physical Quantile	907,871 $\pm$ 96,820		7.20% $\pm$ 1.15%	663,875 $\pm$ 73,821	24,400 $\pm$ 4,106	31.59 $\pm$ 2.61
	Frozen Backbone Direct Quantile MLP	1,680,339 $\pm$ 289,602		0.57% $\pm$ 0.25%	1,660,135 $\pm$ 298,104	2,020 $\pm$ 992	64.38 $\pm$ 12.00
	Global Quantile	1,030,131 $\pm$ 108,592		6.87% $\pm$ 1.39%	785,305 $\pm$ 97,397	24,483 $\pm$ 5,104	36.78 $\pm$ 3.35
	Joint Dense Head	1,020,078 $\pm$ 112,195		6.90% $\pm$ 1.07%	775,899 $\pm$ 96,235	24,418 $\pm$ 4,246	36.35 $\pm$ 3.57
	Joint Routed (Ours)	990,625 $\pm$ 104,487		7.51% $\pm$ 1.22%	720,858 $\pm$ 92,754	26,977 $\pm$ 5,183	35.10 $\pm$ 3.10
	Missingness-Aware GBDT	1,261,941 $\pm$ 112,754		4.88% $\pm$ 0.75%	1,094,730 $\pm$ 120,612	16,721 $\pm$ 3,298	46.62 $\pm$ 3.40

[†]Exceeds nominal 10% violation target ($q^* = 0.90$). Bold numbers denote lowest cost. **Protocol Attribution:** Reports the *Clean-Calibrated Operational Protocol* at $h = 6$ (thresholds frozen on clean validation data); Continuous Physical Quantile collapses to 12.20% violation under Delay-6. For the *State-Conditional Validation-Calibrated Protocol* at immediate horizon ($h = 1$), see Section IV-A2, where clean-calibrated physical rules collapse to 24.0% violation under Delay-6 while state-conditional calibration mitigates violations to 10.7% (physical) and 10.6% (joint model). Test-set iso-multipliers ($\gamma_{iso, test}$) serve strictly as post-hoc diagnostics. [‡]Direct pinball regression on frozen embeddings exhibits severe tail conservatism ($>1.63M$ kW reserve, $<1.2\%$ violation). For calibrated modular baselines (Cascaded Frozen MLP 1.5528M $\pm$ 1.599M vs Joint Routed 16.065M in statistical parity), see Section IV-A3.

TABLE II

SEED-PAIRED DIFFERENCE IN PENALIZED RESERVE-SHORTFALL ENERGY AGAINST JOINT ROUTED ($h = 6$, $\Delta \text{COST} = \text{COST}_{\text{BASELINE}} - \text{COST}_{\text{JOINT ROUTED}}$, POSITIVE INDICATES JOINT ROUTED SAVES ENERGY/COST AT $\rho = 10$).

Regime	Baseline Model	Δ Cost (kW-h)	95% Bootstrap CI	Sig. & Operational Verdict
Clean	Global Quantile	+39,962	[+12,556, +67,367]	Yes ($p < 0.05$)
	Cont. Physical Quantile	-77,659	[-106,175, -49,143]	Yes (Lower Clean)
	Missingness GBDT	+279,576	[+249,539, +309,613]	Yes ($p < 0.0001$)
	Frozen Backbone Direct MLP	+690,695	[+448,769, +932,621]	Yes ($p < 0.01$)
	Joint Dense Head	+32,154	[-1,148, +65,456]	No (Parity, CI crosses 0)
Delay-6	Global Quantile	+17,010	[+3,486, +30,535]	Yes ($p < 0.05$)
	Cont. Physical Quantile	-27,126	[-46,907, -7,346]	Viol. Exceeded (12.20%)
	Missingness GBDT	+156,411	[+140,004, +172,817]	Yes ($p < 0.0001$)
	Frozen Backbone Direct MLP	+518,658	[+307,212, +730,104]	Yes ($p < 0.01$)
	Joint Dense Head	+16,267	[+7,131, +25,402]	Nominal ($p = 0.0251$, non-sig. Bonferroni)
Noise	Global Quantile	+26,475	[+9,121, +43,828]	Yes ($p < 0.05$)
	Cont. Physical Quantile	-28,804	[-47,555, -10,053]	Physical lower cost (7.59% viol., compliant)
	Missingness GBDT	+287,563	[+248,847, +326,279]	Yes ($p < 0.0001$)
	Frozen Backbone Direct MLP	+612,832	[+413,733, +811,931]	Yes ($p < 0.01$)
	Joint Dense Head	+21,066	[+2,419, +39,594]	Nominal ($p < 0.05$, non-sig. Bonferroni)
Markov	Global Quantile	+39,506	[+15,961, +63,052]	Yes ($p < 0.05$)
	Cont. Physical Quantile	-82,754	[-111,026, -54,482]	Yes (Lower Clean)
	Missingness GBDT	+271,316	[+245,044, +297,588]	Yes ($p < 0.0001$)
	Frozen Backbone Direct MLP	+689,714	[+448,089, +930,739]	Yes ($p < 0.01$)
	Joint Dense Head	+29,453	[-2,874, +61,779]	No (Parity, CI crosses 0)

Note: Under Clean and Markov regimes, the 95% bootstrap CI against Joint Dense Head crosses zero, establishing statistical parity resulting from end-to-end task-loss alignment. Nominal routing advantages appear under Delay-6 and Noise, but these gains do not survive Bonferroni multiple-comparison correction ($\alpha = 0.0125$).

primary 589,535 kW · h benchmark, with well-calibrated 7.29% $\pm$ 1.42% violation). Machine-learning representations incur reserve penalties without reliability gains (663,347 kW · h for modular frozen backbones, $+11.8\%$). When telemetry latency increases to $\tau \in [10, 30]$ min with all channels observable, clean-calibrated physical rules collapse (11.18% $\rightarrow$ 13.98% $\rightarrow$ 16.58%). However, state-conditional recalibration fully restores physical compliance (7.1%–7.5% violation) at lower cost than neural models (742,211–939,190 kW · h vs. 786,366–987,760 kW · h), indicating that simple recalibration provides the most cost-effective recovery among evaluated methods under full observability.

- Regime II: Learned Representation Advantage (Unobservable pitch channels / partial observability):** When blade-pitch telemetry is withheld across aggregator or OEM boundaries (no_pitch), physical aerodynamic rules lose direct rotor state awareness. Recalibrated physics based solely on wind-speed binning inflates reserve costs (824,393 kW · h at $\tau = 0$; 928,028 kW · h at $\tau = 10$; 1,010,530 kW · h at $\tau = 20$; 1,101,292 kW · h at $\tau = 30$; 1,378,900 kW · h at $\tau = 60$). In contrast, learned representations (*Frozen Backbone + Residual Quantile*) reconstruct operating states from electromechanical transients (active/reactive power, rotor speed), achieving 755,794 kW · h at $\tau = 0$ (an 8.3% cost reduction of 68,599 kW · h over recalibrated physics) and maintaining consistent cost savings across all latencies (866,753 kW · h at $\tau = 10$, -61.3k; 947,869 kW · h at $\tau = 20$,

-62.7k; 1,054,592 kW · h at $\tau = 30$, -46.7k; 1,330,318 kW · h at $\tau = 60$, -48.6k).

- Regime III: Recalibrated but Tail-Risk Limited ($\tau = 60$ min, Delay-6):** Under 60-minute latency, unadapted physical rules experience systematic collapse (24.02% $\pm$ 1.69% violation, incurring 127.6 MWh of unhedged shortfall exposure). Condition-matching recalibration restores fleet-average violation below the 10.0% target across all three model families (9.46% for physical rules, 9.55% for Frozen Backbone, and 9.62% for Joint Routed), with the recalibrated physical rule delivering the lowest operating cost (1,228,609 kW · h). This directly refutes the prior assumption that multi-step SCADA latency unconditionally collapses all models or makes selective rejection mandatory to meet fleet-average compliance. However, fleet-average compliance conceals critical tail risk: across the five random seeds, every method achieves strictly a 3/5 (60%) seed compliance rate, as Seeds 201 and 203 experience residual breaches (10.2%–10.7%) across all architectures. In this severe telemetry regime, unconstrained neural optimization is vulnerable to tail risk.

TABLE III

MULTIDIMENSIONAL OPERATIONAL PROGRESSION ACROSS TELEMETRY LATENCY $\tau \in [0, 60]$ MIN AND PITCH OBSERVABILITY ($h = 1$, 5-SEED MEAN ON WTB; NOMINAL $\tau = 0$ BASELINE 593,258 kW · h ALIGNS WITHIN 0.6% OF THE PRIMARY FROZEN 589,535 kW · h BENCHMARK). COST IN kW · h ($\rho = 10$).

τ (min)	Pitch	Clean Physical		Recal. Physical		Frozen Backbone		Joint Routed		Operational Regime
		Cost / Viol.	Cost / Viol.	Cost / Viol.	Cost / Viol.	Cost / Viol.	Cost / Viol.			
0	All	593,258 (7.3%)	593,258 (7.3%)	663,347 (6.1%)	663,915 (7.3%)	Regime I (Physics Opt.) Regime II (Repr. Adv.)				
0	None	961,024 (3.2%)	824,393 (5.4%)	755,794 (5.8%)	756,257 (6.4%)					
10	All	753,606 (11.2%) [†]	742,211 (7.4%)	786,366 (6.7%)	789,094 (7.6%)	Regime I (Recal. Suff.) Regime II (Recal. Suff.)				
10	None	1,008,743 (4.8%)	928,028 (6.2%)	866,753 (6.1%)	874,156 (6.6%)					
20	All	896,198 (14.0%) [†]	841,734 (7.1%)	875,838 (7.4%)	879,750 (8.0%)	Regime I (Recal. Suff.) Regime II (Recal. Suff.)				
20	None	1,062,646 (6.1%)	1,010,530 (6.2%)	947,869 (6.7%)	959,934 (7.1%)					
30	All	1,069,266 (16.6%) [†]	939,190 (7.5%)	987,760 (7.1%)	990,091 (7.6%)	Regime I (Recal. Suff.) Regime II (Repr. Adv.)				
30	None	1,138,774 (7.5%)	1,101,292 (6.7%)	1,054,592 (6.4%)	1,064,238 (6.7%)					
60	All	1,656,285 (24.0%) [†]	1,228,609 (9.5%)	1,283,855 (9.6%)	1,274,047 (9.6%)	Regime III (Tail-Risk Limited) Regime II (Repr. Adv.)				
60	None	1,437,280 (11.8%) [†]	1,378,900 (9.0%)	1,330,318 (8.7%)	1,331,840 (8.8%)					

[†]Exceeds nominal 10% violation target ($q^* = 0.90$). Bold indicates lowest compliant operational cost. Under $\tau = 60$ min, by-seed compliance is 3/5 seeds (60%) across models.

To safeguard operations in Regime III under severe communication stress, analogous to wide-area power system controllers where telemetry delays trigger watchdog-governed fallbacks [10] or anomaly mitigations [11], we evaluate a **Selective Decision Abstention (Risk-Coverage Policy)**:

$$r_{i,t}^* = \begin{cases} \hat{r}_{i,t}^{\text{learned}}, & \text{if } U_{i,t} \leq \theta_c \quad (\text{Accepted Prediction}), \\ \hat{r}_{i,t}^{\text{fallback}}, & \text{if } U_{i,t} > \theta_c \quad (\text{Decision Abstention / Refusal}), \end{cases} \quad (14)$$

where $U_{i,t}$ denotes predictive uncertainty (entropy or quantile spread), and θ_c is calibrated on validation data to target coverage

fraction $c \in [0.5, 1.0]$. The fallback reserve $r_{i,t}^{\text{fallback}}$ is defined by the conservative envelope:

$$r_{i,t}^{\text{fallback}} = \max \left(r_{i,t}^{\text{phys, recal}}, [P_{\text{rated}} - \hat{P}_{i,t}]_0^{P_{\text{rated}}} \right), \quad (15)$$

where $[x]_0^{P_{\text{rated}}} = \min(\max(x, 0), P_{\text{rated}})$ clamps available generation headroom against rated nameplate capacity ($P_{\text{rated}} = 1,500$ kW), lower-bounded by the condition-recalibrated aerodynamic quantile rule $r_{i,t}^{\text{phys, recal}}$. To audit this mechanism, Table IV benchmarks selective abstention against two mandatory controls: **Random Abstention** (matching realized empirical coverage $\hat{c}$) and **Uniform Margin Inflation** (evaluated both as an ex-post equal-budget diagnostic benchmark matching total fleet reserve MWh, and as an ex-ante deployable policy with inflation multiplier α_{val} calibrated on validation data).

TABLE IV

RISK-COVERAGE SELECTIVE DECISION ABSTENTION VS. RANDOM AND UNIFORM CONTROLS UNDER DELAY-6 ($\tau = 60$ MIN, $h = 1, 5$ -SEED AGGREGATE ON WTB).

Model	Target c ($\hat{c}$)	Selective Abstention Fleet Viol. / Cost	Random Control Fleet Viol. / Cost	Uniform Inflation Fleet Viol. / Cost
<i>Mode: All Channels Observable</i>				
Joint Routed	1.00 (1.00)	9.62% / 1,274,047	9.62% / 1,274,047	9.62% / 1,274,047
	0.90 (0.94)	9.44% / 1,274,784	9.36% / 1,273,870	9.45% / 1,273,944
	0.80 (0.85)	9.14% / 1,277,532	9.36% / 1,270,990	9.24% / 1,274,781
	0.50 (0.53)	8.54% / 1,276,252	8.78% / 1,265,902	8.80% / 1,277,570
Frozen Backbone	1.00 (1.00)	9.52% / 1,283,268	9.52% / 1,283,268	9.52% / 1,283,268
	0.90 (0.94)	9.30% / 1,277,806	9.36% / 1,281,737	9.36% / 1,283,424
	0.80 (0.85)	9.01% / 1,273,271	9.28% / 1,279,718	9.16% / 1,284,255
	0.50 (0.53)	8.37% / 1,264,384	8.76% / 1,271,385	8.78% / 1,286,221
<i>Mode: No Pitch Telemetry (Withheld)</i>				
Frozen Backbone	1.00 (1.00)	8.68% / 1,331,356	8.68% / 1,331,356	8.68% / 1,331,356
	0.90 (0.92)	8.93% / 1,343,555	8.61% / 1,334,006	8.83% / 1,330,829
	0.80 (0.84)	9.15% / 1,350,059	8.66% / 1,337,746	8.92% / 1,331,452
	0.50 (0.55)	8.84% / 1,371,729	8.55% / 1,353,405	8.29% / 1,333,906

Cost in kW·h ($p = 10$). Fleet target $q^* = 0.90$ (10% viol). Note: In *No Pitch*, uniform margin inflation strictly Pareto-dominates selective abstention: ex-post equal-budget uniform inflation achieves 8.29% violation at 1,334k kW·h, and ex-ante validation-frozen inflation achieves 7.12% violation at 1,348k kW·h (both lower than selective abstention at 8.84% and 1,372k kW·h).

Empirical evaluation reveals that **heuristic uncertainty scores (predictive entropy and quantile spread) fail to isolate tail shortfall risk under severe telemetry staleness**. Under observable telemetry (all), target coverage $c = 0.90$ yields realized empirical coverage $\hat{c} = 93.42\% \pm 0.81\%$, but selective abstention reduces fleet violation only marginally ($9.62\% \rightarrow 9.44\%$ for Joint Routed; $9.52\% \rightarrow 9.30\%$ for Frozen Backbone). This reduction is statistically indistinguishable from random abstention (9.50% and 9.36% , $p > 0.40$), while by-seed compliance remains strictly 3/5 seeds (60%), as Seed 201 accepted violation actually rises to 10.48% (vs. 10.22% unrejected). Under withheld pitch (no_pitch), selective abstention is strictly Pareto-dominated by uniform margin inflation: at $c = 0.50$, selective abstention costs 1,371,729 kW·h (8.84% violation), whereas ex-post test-matched uniform margin inflation achieves lower violation (8.29%) at lower total cost (1,333,906 kW·h, saving 37,823 kW·h), and ex-ante validation-frozen inflation achieves 7.12% violation at 1,348,361 kW·h (saving 23,368 kW·h). Because stale SCADA telemetry corrupts both the point prediction and the uncertainty proxy, selective rejection incurs the penalty of bloated fallback reserves on safe turbines while missing true tail shortfalls. In industrial operations facing multi-step SCADA disruption, **uniform reserve margin inflation and aerodynamic physical fallbacks provide more reliable, cost-effective risk hedging than heuristic selective abstention**.

D. Operating-Boundary Recovery and Point-Forecast Price of Routing

Evaluating point-forecasting accuracy across five WTB seeds, the boundary-forced router achieves an overall RMSE of 236.13 ± 8.41 , compared with 224.34 ± 2.23 for iTransformer and 225.74 ± 2.60 for Graph WaveNet. Train-only inverse-frequency class weighting narrows this difference to 5.59 units (229.93 ± 2.50). This architecture is **deliberately non-RMSE-optimal**: it trades a modest +2.5% whole-sample error margin to preserve operating-regime observability and robust reserve hedging under telemetry degradation. Unconstrained deep baselines minimize squared error across prevailing stationary conditions, treating deviations near rated wind speed identically to low-wind fluctuations. Under telemetry staleness, this indifference to operating regimes triggers severe tail shortfalls (78.3–82.5 MWh unhedged shortfall exposure, whereas the boundary-aware representation restricts exposure to 53.9 MWh, achieving an approximately 31%–35% shortage reduction). In parallel, the boundary-guided model preserves physical regime alignment (0.8716 ± 0.0418 NMI and 0.9166 ± 0.0371 ARI under strict masking; 0.721 NMI under train-only supervision). As illustrated in the 2D Pareto frontier in **Supplementary Figure S1**, the learned representation doubles transition regime recall (0.416 vs. 0.196 for degraded physics) while limiting

Delay-6 shortage exposure to 53.9 MWh, indicating that pre-dispatch reliability on this benchmark prioritizes boundary-state tracking over marginal whole-sample RMSE gains.

Input ablation stress guards verify that routing fidelity does not arise from target leakage: withholding Patv or Pab_mean preserves mean NMI at 0.881 and 0.878, while lagging Patv and lagging pitch/wind yield 0.763 and 0.684. In addition, gate matching to the active operating regime peaks at the transition step (0.815) and attenuates under three- and six-step lead shifts (0.361 and 0.405), indicating that gate activations track dynamic physical state transitions.

E. Withheld-Channel Electromechanical Signature Recovery

The operational integrity of the blind-spot fallback rests on whether the model can infer operational regime boundaries when primary pitch and anemometer channels are withheld across commercial OEM protocol boundaries or VPP administrative interfaces. To test whether secondary electromechanical consequence channels carry sufficient information to reconstruct operating state posteriors without circular dependence, we formulate a strict **counterfactual stress-testing probe**: withholding Wspd and Pab_mean while retaining active power and downstream electromechanical channels yields a mean NMI of 0.561 and ARI of 0.635 across five seeds on WTB (Table V). Removing active power (signature_core) drops mean NMI to 0.367, whereas permuted label negative controls collapse to 4.0×10^{-6} , ruling out chance alignment under permuted controls.

TABLE V

SIGNATURE-GATE IDENTIFIABILITY PROBE ACROSS COMMERCIAL WIND FARMS UNDER HETEROGENEOUS PITCH OBSERVABILITY. DEFINING CHANNELS ARE WITHHELD FROM THE GATE ANCHOR; ALL VALUES ARE MEAN $\pm$ SD OVER 5 SEEDS.

Probe	RMSE	NMI / ARI	Interpretation
Canonical full-anchor	229.93	0.7208 / 0.7398	Train-only clean baseline
signature_full (no Wspd/Pab)	241.84 $\pm$ 7.19	0.5613 $\pm$ 0.0199	Boundary recoverable from conseq. channels
signature_core (no Wspd/Pab/Patv)	302.10 $\pm$ 9.16	0.3671 $\pm$ 0.0372	Weaker non-power signature; 3/5 seeds collapsed
sig_full_shuffled	241.50 $\pm$ 10.25	4.0e-6 $\pm$ 2.1e-6	Negative control: chance level under permuted labels
Unconstrained MoE	—	0.0147 / —	No declared-boundary supervision
<i>Cross-farm replication, ENGIE La Haute Borne (4 turbines, 99% pitch)</i>			
LHB canonical (full anchor)	185.0 $\pm$ 1.9	0.9752 $\pm$ 0.0088	Full-anchor baseline, identical protocol
LHB signature_full	185.1 $\pm$ 1.4	0.6743 $\pm$ 0.1209	Boundary signature transfers; min-seed 0.499
LHB signature_core	188.0 $\pm$ 1.0	0.5750 $\pm$ 0.0423	Non-power signature transfers; 0/5 collapsed
<i>Commercial UK wind plants (Penmanshiel MM82, Kelmarsh MM92)</i>			
Penmanshiel signature_full	—	0.3024 $\pm$ 0.0343	Above the 0.20 criterion; shuffled 4.4e-5
Penmanshiel signature_core	—	0.1954 $\pm$ 0.0961	Borderline: below 0.20 but 4400x above shuffled chance
Kelmarsh signature_full	—	0.3402 $\pm$ 0.0736	Above the 0.20 criterion; shuffled 9.2e-5
Kelmarsh signature_core	—	0.3775 $\pm$ 0.0783	Above the 0.20 criterion

To prevent over-generalization and evaluate external deployment boundaries, we strictly disentangle three disparate lines of external empirical evidence across commercial plants:

- Within-plant local retraining (ENGIE La Haute Borne & UK dual-farm replication):** Models trained and evaluated locally under site-specific rated aerodynamic parameters demonstrate that the neural architecture adapts to distinct turbine technologies, confirming that consequence signatures are recoverable from electromechanical transients without direct pitch observation:
 - ENGIE La Haute Borne (4 turbines, 99% pitch, $v_{\text{rated}} = 14.5 \text{ m s}^{-1}$):** Withheld-channel probe retains an NMI of 0.674 for signature_full and 0.575 for signature_core (collapsing to 1.92×10^{-4} under permuted controls; five-seed routing NMI 0.941, ARI 0.971).
 - Kelmarsh (6 Senvion MM92 turbines, $v_{\text{rated}} = 12.5 \text{ m s}^{-1}$):** With defining wind-speed and pitch-angle channels withheld, signature_full achieves an NMI of 0.3402 ± 0.0736 (3700x above shuffled chance). In 5-seed local chronological dispatch replay at $h = 6$, locally retrained Joint Routed yields $70,375 \pm 16,324 \text{ kW} \cdot \text{h}$ (Clean) and $109,359 \pm 13,634 \text{ kW} \cdot \text{h}$ (Delay-6), outperforming GBDT (+44,891 kW·h and +29,621 kW·h, $p < 0.05$) and continuous physical quantiles (82,456 kW·h and 116,409 kW·h).
 - Penmanshiel (15 Senvion MM82 turbines, $v_{\text{rated}} = 14.5 \text{ m s}^{-1}$):** With defining channels withheld, signature_full achieves

0.3024 ± 0.0343 (6800x above shuffled chance). In local dispatch replay at $h = 6$, locally retrained Joint Routed yields $831,251 \pm 112,331$ kW · h (Clean) and $813,831 \pm 102,946$ kW · h (Delay-6), versus GBDT ($1,064,962$ kW · h and $1,086,102$ kW · h, $p < 0.001$), where GBDT experiences severe violation inflation (24.7% to 26.5%). These empirical findings demonstrate neural architectural adaptability to site-specific aerodynamic rated parameters under local retraining, while bounding cross-farm generalizability to localized recalibration.

2. **Directional transfer boundaries across disparate wind plants:** In contrast to local chronological retraining, direct zero-shot cross-farm transfer without site-specific recalibration reveals pronounced directional transfer sensitivity. As audited by our pre-registered external wind guard (`external_wind_guard.json`), transferring representations from Kelmarsh to Penmanshiel preserves moderate boundary alignment (NMI 0.752–0.770), whereas reverse transfer from Penmanshiel to Kelmarsh yields NMI 0.341–0.505 (pooled bidirectional mean NMI 0.557). This directional asymmetry reflects that aerodynamic operating boundaries are coupled to turbine rotor dimensions and local wake topographies, reinforcing that reliable multi-farm deployment requires site-specific sensor calibration and local retraining.
3. **Longitudinal multi-year operational stability:** Across decadal longitudinal archives (Kelmarsh, 9 years; Penmanshiel, 8.6 years), walk-forward evaluations examine periodic quantile recalibration under multi-year climatological drift and turbine component aging. These longitudinal evaluations demonstrate the operational mechanics of periodic schedule recalibration across multi-year operating horizons, establishing empirical baselines for long-term fleet management under environmental drift.

F. Degradation Resilience and Early Warning Dynamics

Table VI evaluates detection fidelity under fair input degradation (Unified Arrival Layer), wherein full-stream history shifts and anchor delays are applied symmetrically to all models. Under clean telemetry, deterministic physical thresholds exhibit higher sensitivity (recall 1.000 vs. 0.9705), while logistic regression achieves 1.000 recall with 0.879 precision (vs. 0.630 for the routed posterior). Under 6-step latency ($d = 6$, 60-min history stall), however, deterministic rule recall collapses from 1.000 to 0.196 (0.249 F1). In contrast, the spatio-temporal boundary posterior sustains an early-window recall of 0.4170 ± 0.1137 (+0.221 gain, more than doubling detection sensitivity) with an F1 score of 0.3802 (+0.131 gain). While 60-minute communication latency naturally attenuates absolute detection recall ($0.971 \rightarrow 0.417$), the dynamic wake graph doubles surviving operational resilience without look-ahead leakage.

TABLE VI

EARLY MPPT-TO-PITCH DETECTION UNDER FAIR INPUT DEGRADATION (UNIFIED ARRIVAL LAYER) ACROSS 5 TRAIN-ONLY SEEDS. FULL-STREAM HISTORY SHIFT AND ANCHOR DELAY APPLIED SYMMETRICALLY TO GATE AND RULE.

Scenario	Setting	Gate	Rule	Gain
Clean anchors	—	0.9705 ± 0.0299	1.000	-0.029
Delay (fair)	1 step	0.8346 ± 0.0771	0.655	+0.179
Delay (fair)	3 steps	0.6200 ± 0.1209	0.381	+0.239
Delay (fair)	6 steps	0.4170 ± 0.1137	0.196	+0.221
Noise (fair)	Wspd 0.5, Pab 1.0	0.9635 ± 0.0375	0.811 ± 0.016	+0.152
Noise (fair)	Wspd 1.0, Pab 2.0	0.9505 ± 0.0363	0.680 ± 0.019	+0.271
Availability	75% labels	0.971	0.744	+0.227
Availability	50% labels	0.971	0.508	+0.462
Availability	25% labels	0.971	0.242	+0.729

Under industrial two-state Markov-Gilbert burst dropouts ($p_{GB} = 0.08$, $p_{BB} = 0.75$, $d \leq 6$) evaluated symmetrically across both anchor and encoder history channels, the stale threshold rule’s recall drops to 0.840 with an F1 of 0.904, whereas the jointly-learned routed posterior sustains a recall of 0.962 ± 0.023 and an F1 of 0.925 ± 0.042 during bursts (mean F1 gain +0.021, 95% bootstrap CI $[-0.014, +0.050]$). During dynamic transitions, the spatio-temporal graph provides vital defense-in-depth against communication dropouts.

G. Boundary-Window Reserve Vignette and Benchmark Comparison

At a shortage-to-reserve cost ratio $\rho = 10$ on the transition boundary band (aggregated over the 35-day test set), the boundary-router gate-bin policy costs 95.13M, outperforming the same-model global rule (99.55M), while the low-RMSE Graph WaveNet with physical-bin reference achieves 84.31M (corresponding to ≈ 1.28 M kW · h per single dispatch interval). This translates into approximately 637 avoided

MWh-equivalent shortage cells and a 442k illustrative reserve-cost-scale marker at 100 EUR/MWh.

H. Point of Common Coupling (PCC) Portfolio Smoothing & Exploratory Longitudinal Drift Analysis

Aggregating decentralized turbine power flows at the Point of Common Coupling (PCC) bus ($P_{\text{farm}} = \sum_{i \in \mathcal{G}} P_{i,t}$, averaging ~ 121 active turbines) unlocks spatial portfolio diversification: localized high-frequency aerodynamic turbulence cancels out across the array, leaving boundary transitions as the dominant source of reserve shortfall risk. Across the full operational envelope, joint posterior aggregate quantile pricing reduces reserve cost by -2.32 M kWh (95% bootstrap CI $[-6.07\text{M}, 1.73\text{M}]$) against global PCC quantiles and by -1.37 M kWh (CI $[-6.33\text{M}, 2.82\text{M}]$) against Gaussian parametric sizing; in transitional regimes (10%–90% pitching), it saves -1.33 M kWh (CI $[-1.68\text{M}, -1.07\text{M}]$, strictly excluding zero) over continuous physical pitch, confirming net risk-reduction value after fleet-wide spatial cancellation. Interfacing scheduled reserves with a co-located battery energy storage system (BESS, 20, MWh / 10, MW) under receding-horizon Model Predictive Control (MPC) reduces Delay-6 dispatch cost by 32.1% ($\$487,746$ to $\$331,257$) and cuts shortage energy by 45.7% (2,514.9 to 1,401.2, MWh; Supplementary Table A11-BESS). These upstream values represent the PSREI risk proxy ($\rho = 10$) rather than wholesale market cashflow settlements.

Across two European commercial wind plants with longitudinal records (Kelmarsh, 9 years, 2016–2024, 6 MM92 turbines; Penmanshiel, 8.6 years, 2016–2024, 15 MM82 turbines), exploratory evaluations examine periodic quantile recalibration under multi-year climatological drift and turbine aging. These longitudinal evaluations illustrate the operational mechanics of periodic schedule recalibration across decadal operating horizons, establishing empirical baselines for long-term fleet management.

V. DISCUSSION

A. Operational Diagnostics and Deployment Boundaries

The boundary-aware architecture functions as a targeted diagnostic instrument for pre-dispatch reserve risk screening, prioritizing transition-window tail reliability over whole-sample point accuracy. When bulk energy dispatch operates under pristine SCADA telemetry, unconstrained spatio-temporal architectures (Graph WaveNet, iTransformer) deliver higher fleet-wide point accuracy; conversely, when telemetry is delayed, corrupted, or pitch-withheld, boundary-risk posteriors provide indispensable diagnostic defense against tail shortfalls. To govern industrial deployment, we delineate three empirical validation axes:

1. **Directional transfer boundaries across disparate wind plants:** Direct zero-shot cross-farm transfer exhibits pronounced directional asymmetry (Kelmarsh $\rightarrow$ Penmanshiel NMI 0.752–0.770, ARI 0.735–0.761 vs. Penmanshiel $\rightarrow$ Kelmarsh NMI 0.341–0.505, pooled bidirectional mean NMI 0.557; `external_wind_guard.json`). Unadapted zero-shot transfer is strictly circumscribed by turbine rotor dimensions and local array aerodynamic geometry. Reliable multi-site transfer requires site-specific sensor recalibration.
2. **Within-plant local retraining:** Chronological retraining across ENGIE La Haute Borne (five-seed routing NMI 0.941, ARI 0.971; canonical NMI 0.975; withheld probe 0.674/0.575), Kelmarsh (probe 0.340/0.378), and Penmanshiel (probe 0.302, `signature_core` 0.195; Supplementary Table A9d) confirms neural architectural adaptability to site-specific rated wind speeds and turbine technologies.
3. **Longitudinal rolling walk-forward evaluation:** Across decadal longitudinal archives (Kelmarsh, 9 years; Penmanshiel, 8.6 years), periodic 2-year sliding window evaluations characterize operational recalibration under climatological drift, tracking parameter evolution over decadal operating horizons.

External walk-forward rolling evaluation establishes empirical physical boundary conditions for the pre-registered admission protocol (Supplementary Table A14). On ENGIE La Haute Borne (4 turbines, 99% complete pitch), walk-forward pooled quarterly evaluation (Q1–Q3) incurs a positive cost delta of +1.01 M kWh (95% bootstrap CI $[+0.29\text{M}, +1.76\text{M}]$, strictly excluding zero) versus the global quantile and +0.92 M kWh (CI $[+0.26\text{M}, +1.61\text{M}]$) versus continuous physical pitch (Supplementary Table A11h). This defines a clear operational boundary condition: on a micro-scale 4-turbine site situated

in complex rolling terrain, spatial wake propagation lacks multi-path physical redundancy, causing spatial graph convolution to overfit localized terrain distortions and amplify variance during seasonal regime shifts (+968.1k kWh in Q3, while Q1–Q2 cross zero). Expanding calibration to a 180-day annual window compresses the cost gap to +43k kWh (CI [-28.7k, +141.2k], strictly crossing zero), bounding learned spatio-temporal representations to utility-scale turbine arrays (≥ 6 –15 turbines) and degraded/pitch-sparse telemetry.

Evaluating across shortage penalty ratios $\rho \in \{5, 10, 20\}$ articulates an operational envelope (“Telemetry Availability $\times$ Penalty Ladder”). At $\rho = 10$, learned posteriors dominate unconditioned baselines. At $\rho = 20$, asymmetry emerges: at Kelmarsh, physical rules recover and surpass soft-gate pricing by +0.95M kWh (CI [+0.15M, +1.65M]); at Penmanshiel, the soft gate leads physical rules by -7.51M kWh (CI [-13.51M, -1.25M]) while crossing zero versus the unconditioned global baseline (-5.15M kWh, CI [-12.47M, +2.17M]). Direct reserve deployment at new wind plants requires pitch or proxy observability, boundary support, compatible geometry, local recalibration, and a held-out validation audit [8].

VII. LIMITATIONS

WTB pseudo-labels derive from anemometer wind speed and pitch angle, channels present in full gate anchors. Counterfactual withheld-channel stress testing indicates that the boundary survives without direct pitch observations via consequence signatures across four commercial plants (permuted controls at chance). Removing pitch dispersion retains non-power NMI 0.309 on WTB and 0.578 on La Haute Borne, verifying that dispersion does not carry the signal. Operational regime identification operates as an anchor-constrained diagnostic grounded in engineering control boundaries, rather than unsupervised physical law discovery [18], [25]. Input features include observation history $\text{Pat}v_{t-H+1:t}$ and dispatch-time $\text{Pat}v_t$, with target trajectories commencing strictly at $t + 1$. Expert semantics reflect the declared aerodynamic partition; unassigned logits do not represent universal physical states.

The analytical scope of this study is formally bounded to Level 1 local pre-dispatch risk screening proxies (PSREI at $\rho = 10$) designed to mitigate unhedged energy imbalances entering real-time grid balancing. The formulation abstracts away Level 2 transmission network clearing with Locational Marginal Prices (LMPs), full AC-OPF power-flow constraints, and multi-settlement cashflows [27], [29]. Core empirical findings remain five-seed WTB recovery, mechanism ablation, La Haute Borne local replication, and reserve risk triage.

VIII. CONCLUSION

This study establishes an empirical operational boundary and value-of-information framework delineating the complementary roles of aerodynamic power rules, state-conditional recalibration, and learned representations under SCADA telemetry degradation. Across a 5-seed benchmark on the 134-turbine WTB plant, continuous aerodynamic quantile rules achieve the lowest reserve-screening surrogate cost under pristine telemetry (589,535 kW $\cdot$ h at $h = 1$, 881,367 kW $\cdot$ h at $h = 6$), but undergo systematic reliability collapse under a 6-step (60 min) severe outage-envelope stress condition, surging violation rates to $24.0\% \pm 1.7\%$ ($h = 1$) and $12.20\% \pm 1.25\%$ ($h = 6$). Under Delay-6, Continuous Physical Quantile reaches 1,136,466 $\pm$ 84,101 kW $\cdot$ h at $h = 6$ and 1,228,609 $\pm$ 71,940 kW $\cdot$ h at $h = 1$ under condition-matching recalibration.

Under frozen validation calibration, state-conditional recalibration of physical rules recovers violation to 10.7% and cuts shortage from 127.6 MWh to 57.0 MWh (55.3% reduction) at lower cost than neural models, achieving the lowest evaluated surrogate cost under full observability. When blade-pitch registers are unobservable across aggregator or OEM firewalls, learned representations reconstruct operating regimes from secondary electromechanical transients, consistently outperforming recalibrated physics by 46.7k–68.6k kW $\cdot$ h across all latencies and doubling regime recall (0.416 vs. 0.196 under Delay-6). The boundary-aware representation is deliberately non-RMSE-optimal, accepting a modest 2.5% point-accuracy margin to reduce unhedged tail shortfall exposure by approximately 31%–35% (78.3–82.5 MWh $\rightarrow$ 53.9 MWh) compared to unconstrained deep architectures. Capacity-matched Dense and MoE heads achieve statistical parity under Clean telemetry (cost ratio 1.045, $p = 0.380$) with a modest Delay-6 margin (ratio 1.056, uncorrected $p = 0.038$), while modular architectures match tail reliability (9.7% violation),

indicating that operational value on this benchmark originates from shared spatio-temporal representations and residual quantile calibration rather than dynamic routing. Direct zero-shot cross-farm transfer exhibits directional asymmetry (Kelmarsh $\rightarrow$ Penmanshiel NMI 0.752–0.770 vs. Penmanshiel $\rightarrow$ Kelmarsh 0.341–0.505, pooled 0.557), bounding reliable deployment to sites with local retraining (ENGIE La Haute Borne, NMI = 0.941). These empirical findings position machine learning as an accountable, boundary-aware diagnostic instrument for resilient wind energy operations, operating alongside calibrated physical curves under verifiable communication boundaries.

VIII. AI USE STATEMENT

The authors used OpenAI ChatGPT/Codex only for language editing, consistency checks, and submission-material drafting; all data, analyses, references, conclusions, and final text were reviewed and controlled by the authors.

IX. CODE AND DATA AVAILABILITY

The raw KDD Cup 2022, ENGIE La Haute Borne, Kelmarsh, and Penmanshiel SCADA datasets are public (raw third-party data not redistributed). Code, configurations, releasable derived tables, figure data, and checkpoints will be made available with the article. The reproduction package covers WTB routing, reserve audit, anchor-stress caches, early-warning label degradation, classifier control, class-weight sensitivity, external diagnostics, and evidence-freeze protocols across declared seeds. Results are statistically reproducible across declared seeds rather than bitwise deterministic across all GPU/CUDA environments. The analysis involves no human subjects.

REFERENCES

- [1] J. Dowell and P. Pinson, “Very-short-term probabilistic forecasting by sparse vector autoregression,” *IEEE Transactions on Smart Grid*, vol. 7, no. 2, pp. 1063–1070, 2016, doi: 10.1109/TSNG.2015.2424078.
- [2] J. Kuse, E. Cramer, B. Schuler, and D. Wittmann, “Physics-informed machine learning for power grid frequency modeling,” *PRX Energy*, vol. 2, no. 4, p. 043003, 2023, doi: 10.1103/PRXEnergy.2023.043003.
- [3] P. Pinson, “Wind energy: Forecasting challenges for its operational management,” *Statistical Science*, vol. 28, no. 4, pp. 564–585, 2013, doi: 10.1214/13-SS045.
- [4] R. Doherty and M. O’Malley, “A new approach to quantify reserve demand in systems with significant installed wind capacity,” *IEEE Transactions on Power Systems*, vol. 20, no. 2, pp. 587–595, 2005, doi: 10.1109/TPWRS.2005.846206.
- [5] Y. Wang, F. Zhang, H. Kou, R. Zou, Q. Hu, and J. Wang, “A review of predictive uncertainty modeling techniques and evaluation metrics in probabilistic wind speed and wind power forecasting,” *Applied Energy*, vol. 396, p. 126234, 2025, doi: 10.1016/j.apenergy.2025.126234.
- [6] J. G. Slootweg, S. W. H. de Haan, H. Polinder, and W. L. Kling, “General model for representing variable speed wind turbines in power system dynamics simulations,” *IEEE Transactions on Power Systems*, vol. 18, no. 1, pp. 144–151, 2003, doi: 10.1109/TPWRS.2002.807113.
- [7] E. Gaertner et al., “Definition of the IEA wind 15-megawatt offshore reference wind turbine,” National Renewable Energy Laboratory (NREL), Golden, CO, NREL/TP-5000-76008, 2020, doi: 10.1171/1603478.
- [8] J. Tautz-Weinert and S. J. Watson, “Using SCADA data for wind turbine condition monitoring: A review,” *IEEE Renewable Power Generation*, vol. 11, no. 4, pp. 382–394, 2017, doi: 10.1049/rpg.2016.0248.
- [9] M. A. Ullah, K. Mishra, and H. Alves, “Enabling mVTC in remote areas: LoRaWAN and LEO satellite integration for offshore wind farm monitoring,” *IEEE Transactions on Industrial Informatics*, vol. 18, no. 6, pp. 3744–3753, 2022, doi: 10.1109/TII.2021.312386.
- [10] B. J. Pierre et al., “Design of the Pacific DC Inertial wide area damping controller,” *IEEE Transactions on Power Systems*, vol. 34, no. 5, pp. 3594–3604, 2019, doi: 10.1109/TPWRS.2019.2903782.
- [11] G. Ravikumar and M. Govindarasu, “Anomaly detection and mitigation for wide-area damping control using machine learning,” *IEEE Transactions on Smart Grid*, vol. 15, no. 6, pp. 5950–5951, 2024, doi: 10.1109/SG.2020.2993513.
- [12] S. Daens, T. Verstraeten, P.-J. Daems, A. Nowé, and J. Helsen, “Spatio-temporal graph neural networks for power prediction in offshore wind farms using SCADA data,” *Wind Energy Science*, vol. 10, no. 6, pp. 1157–1152, 2025, doi: 10.5194/wes-10-1157-2025.
- [13] Z. Wu, S. Pan, G. Long, and C. Zhang, “Graph WaveNet for deep spatial-temporal graph modeling,” in *Proceedings of the 28th international joint conference on artificial intelligence (IJCAI)*, 2019, pp. 1907–1913, doi: 10.2496/3ijcai.2019/264.
- [14] S. Guo, Y. Lin, N. Feng, C. Song, and H. Wan, “Attention based spatial-temporal graph convolutional networks for traffic flow forecasting,” in *Proceedings of the AAAI conference on artificial intelligence*, 2019, pp. 922–929, doi: 10.1609/aaai.v33i01.3301998.
- [15] L. Bai, Y. Yao, C. Li, X. Wang, and C. Wang, “Adaptive graph convolutional recurrent network for traffic forecasting,” in *Advances in neural information processing systems*, 2020, pp. 17804–17815.
- [16] J. Park and J. Park, “Physics-induced graph neural network: An application to wind-farm power estimation,” *Energy*, vol. 187, p. 115883, 2019, doi: 10.1016/j.energy.2019.115883.
- [17] D. Kim, G. Ryu, and C. Moon, “Accuracy of a short-term wind power forecasting model based on deep learning using LiDAR-SCADA integration: A case study of the 400-MW Ashfall offshore wind farm,” *Applied Energy*, vol. 373, p. 123882, 2024, doi: 10.1016/j.apenergy.2024.123882.
- [18] N. Zehabiyah-Rezaie, A. Jostlidis, and M. Abkar, “Physics-guided machine learning for wind-farm power prediction: Toward interpretability and generalizability,” *PRX Energy*, vol. 2, no. 1, p. 013009, 2023, doi: 10.1103/PRXEnergy.2013009.
- [19] E. A. Bossanyi, “The design of closed loop controllers for wind turbines,” *Wind Energy*, vol. 3, no. 3, pp. 149–163, 2000, doi: 10.1002/we.34.
- [20] F. D. Bianchi, H. De Battista, and R. J. Mantz, *Wind turbine control systems: Principles, modelling and gain scheduling design*. London: Springer, 2006, doi: 10.1007/1-84628-493-7.
- [21] L. Y. Pao and K. E. Johnson, “Control of wind turbines: Approaches, challenges, and recent developments,” *IEEE Control Systems Magazine*, vol. 31, no. 2, pp. 44–62, 2011, doi: 10.1109/MCS.2010.939662.
- [22] N. Shazeer et al., “Outrageously large neural networks: The sparsely-gated mixture-of-experts layer,” in *International conference on learning representations*, 2017.
- [23] W. Fedus, B. Zoph, and N. Shazeer, “Switch transformers: Scaling to trillion parameter models with simple and efficient sparsity,” *Journal of Machine Learning Research*, vol. 23, pp. 1–39, 2022.
- [24] A. Karpatne et al., “Theory-guided data science: A new paradigm for scientific discovery from data,” *IEEE Transactions on Knowledge and Data Engineering*, vol. 29, no. 10, pp. 2318–2331, 2017, doi: 10.1109/TKDE.2017.2720168.
- [25] G. E. Karniadakis, J. G. Kevrekidis, L. Lu, P. Perdikaris, S. Wang, and L. Yang, “Physics-informed machine learning,” *Nature Reviews Physics*, vol. 3, no. 6, pp. 422–440, 2021, doi: 10.1038/s42254-021-00314-5.
- [26] E. Elia, M. Milligan, and B. Kirby, “Operating reserves and variable generation,” National Renewable Energy Laboratory, Golden, CO, NREL/TP-5000-51978, 2011, doi: 10.2172/1023095.
- [27] J. B. Bremes, “Probabilistic wind power forecasts using local quantile regression,” *Wind Energy*, vol. 7, no. 1, pp. 47–54, 2004, doi: 10.1002/we.107.
- [28] Y. Zhang, J. Wang, and X. Wang, “Review on probabilistic forecasting of wind power generation,” *Renewable and Sustainable Energy Reviews*, vol. 32, pp. 255–270, 2014, doi: 10.1016/j.rser.2014.01.033.
- [29] Z. Zhou et al., “Application of probabilistic wind power forecasting in electricity markets,” *Wind Energy*, vol. 16, no. 3, pp. 321–338, 2013, doi: 10.1002/we.1496.
- [30] L. Li, Y. C. Shahabi, and Y. Liu, “Diffusion convolutional recurrent neural network: Data-driven traffic forecasting,” in *International conference on learning representations*, 2018.
- [31] H. Hersbach et al., “The ERA5 global reanalysis,” *Quarterly Journal of the Royal Meteorological Society*, vol. 146, no. 730, pp. 1999–2049, 2020, doi: 10.1002/qj.3803.
- [32] H. A. Nielsen, H. Madsen, and T. S. Nielsen, “Using quantile regression to extend an existing wind power forecasting system with probabilistic forecasts,” *Wind Energy*, vol. 9, no. 1–2, pp. 95–108, 2006, doi: 10.1002/we.180.
- [33] J. Zhou et al., “SDWPF: A dataset for spatial dynamic wind power forecasting over a large turbine array,” *Scientific Data*, vol. 11, p. 649, 2024, doi: 10.1038/s41597-024-03472-5.
- [34] Y. Nie, N. H. Nguyen, P. Sinthong, and J. Kalagnanam, “A time series is worth 64 words: Long-term forecasting with transformers,” in *International conference on learning representations*, 2023.
- [35] Y. Liu et al., “Transformer: Inverted transformers are effective for time series forecasting,” in *International conference on learning representations*, 2024.
- [36] A. Das, W. Kong, A. Leach, and R. Sen, “Long-term forecasting with TIDE: Time-series dense encoder,” in *Transactions on machine learning research*, 2023.